

INBRIEF



Sanjay Agrawal sworn in as Rajasthan HC Chief Justice

Justice Sanjay Kumar Agrawal was sworn in as the 44th Chief Justice of the Rajasthan High Court at a ceremony held at Lok Bhavan in Jaipur on Monday. Governor Haribhau Bagade administered the oath of office in the presence of Chief Minister Bhajan Lal Sharma, his Cabinet colleagues, judges and administrative officials. The Union government notified Justice Agrawal's elevation as the Chief Justice and appointment in Rajasthan following a recommendation by the Supreme Court Collegium. He was first appointed as an additional judge of the Chhattisgarh High Court in September 2013.

HC sets aside arrest warrant against TMC's Mahua Moitra

The Calcutta High Court on Monday set aside an arrest warrant issued against TMC MP Mahua Moitra by a Krishnanagar court, and ordered transfer of the case against her over alleged hate speech to the special MP/MLA court at Bidhannagar here. Ms. Moitra, in a petition before the High Court, had challenged the legality of the arrest warrant issued against her by the Krishnanagar court. The High Court had on August 24 stayed the arrest warrant, which was issued by the third judicial magistrate at Krishnanagar due to Ms. Moitra's alleged non-compliance with directions to appear before it.

‘Conduct worthy of derision’: HC pulls up District Magistrate over NSA detention of student

Ishita Mishra
NEW DELHI

The Allahabad High Court has come down heavily on Gautam Buddha Nagar District Magistrate Medha Roopam, who had issued the order to detain a 25-year-old Delhi University law student under the National Security Act (NSA) over a workers' protest in Noida in April this year. In a detailed order released on Monday, the court observed that Ms. Roopam “desired to make an example” out of the petitioner. A Bench of Justices Atul Sreedharan and Achal Sachdev had on September 2 quashed the NSA detention of Akriti Chaudhary, who had spent nearly five months in custody. The Bench had allowed Ms. Chaudhary's habeas corpus petition and ordered her immediate re-



Gautam Buddha Nagar District Magistrate Medha Roopam

lease, unless she was required in another case. The court flagged discrepancies in the arrest notice and termed the State's version a “concocted story”.

‘Guilty of violating oath’ Pulling up the District Magistrate, the Bench noted, “In this case, the conduct of the District Magistrate, who passed the impugned order, is worthy of derision.” The court observed that the District Magistrate “desired to set an exam-

Shinde suspends Sena leader arrested over attack on medical staff

Sena chief says such incidents will not be tolerated; police on the lookout for remaining accused; second such ‘assault’ by a Sena leader since July 6

Yohaán Varghese
MUMBAI

Maharashtra Deputy Chief Minister and Shiv Sena chief Eknath Shinde on Monday suspended the party's Palghar city chief Rahul Gharat following the latter's arrest over the alleged assault on employees of a private hospital by Shiv Sena workers a day ago. Mr. Shinde said such incidents would not be tolerated and warned of strict action against anyone found responsible.

Inspector Kishor Mahanubhav of the Palghar police station said Mr. Gharat was produced before a local magistrate, who remanded him to police custody for a day.

This is the second such incident of assault by a Sena leader in the recent past. Sena corporator Ramesh Mhatre was arrested for allegedly assaulting doctors at a Thane hospital on July 6.

A wider probe into the Palghar attack was under way even as the police re-



Maharashtra Deputy CM and Shiv Sena chief Eknath Shinde warned of strict action against anyone found responsible. FILE PHOTO

main on the lookout for the remaining accused persons, the officer said.

According to the police, five ‘govindas’ who were a part of a Shiv Sena Dahi Handi celebrations in Palghar were admitted to Relief Hospital in Palghar.

“The bill for the treatment of the govindas came to ₹19,000. Seventeen Sena members came to the hospital and had already paid ₹10,000; the remaining amount was yet to be paid. One of the ‘govindas’, aged 19, wanted to go home in Virar. But the doctor said he wanted to keep him for 24-

hour observation,” Mr. Mahanubhav said. The officer added that the party members refused to pay the remaining amount until the 19-year-old was released, following which an argument broke out and the party workers assaulted the staff.

CCTV footage Mr. Gharat, who was part of the group, allegedly slapped a staff member. CCTV footage from the hospital purportedly shows one person slapping a receptionist. The police have registered an FIR against 17 individuals.

HC warns Bengal govt. of passing blanket order protecting Abhishek

Seniuti Sengupta
KOLKATA

The Calcutta High Court on Monday warned the State government of passing a blanket order in favour of Trinamool Congress (TMC) general secretary Abhishek Banerjee, restraining the registration of any further FIRs against him. The court also provided interim protection to Mr. Banerjee till November 30 in three FIRs linked to alleged irregularities in the TMC's Sebaash-ray health camps.

“Enough is enough. I have been hearing from May 4. Now I am going to pass orders relying on the one passed in Suwendu Adhikari's case when he was Leader of Opposition,” the Bench of Justice Saugata Bhattacharyya stated.

The court directed the police not to take any coercive action against Mr. Banerjee till November 30

and also asked the Diamond Harbour MP to co-operate in the investigation. Appearing for Mr. Banerjee, senior advocate Gopal Sankaranarayanan, told the court that seven out of 16 FIRs against Mr. Banerjee has been filed by his unsuccessful opponent in the general election, Abhijit Das.

‘Akin to Suwendu case’ Mr. Sankaranarayanan submitted that Mr. Banerjee was facing a situation similar to that considered by the High Court in proceedings concerning Mr. Adhikari. “In Suwendu Adhikari's case, we have the exact same prayer. In our earlier plea your Lordship granted us relief in May and June. After those two orders, these FIRs were filed, exactly like in Mr. Adhikari's case, trying to circumvent the protection we have,” he submitted.

BJP to begin 4-day Statewide stir in Jharkhand over cases against students from today

Amit Bhelari
PATNA

The Opposition Bharatiya Janata Party (BJP) in Jharkhand has finalised preparations for a Statewide agitation starting September 8 to demand the withdrawal of cases filed by the police against students protesting against alleged irregularities in recruitment exams, including Jharkhand Public Service Commission (JPSC) and Jharkhand Staff Selection Commission (JSSC).

Responsibility assigned The party has assigned responsibilities to workers and asked prominent leaders to join the protests in their respective areas.

State general secretary Amar Kumar Bauri on Monday said that a four-day agitation, protests and the gherao of Deputy Commissioner and Superintendent of Police offices will be conducted at 24 district headquarters on a scheduled basis from September 8 to 11.

“Following the directives of State party president, specific leaders have been entrusted with the responsibility of leading the movement in different



Police dispersing student protesters attempting to breach barricades during an Assembly march in Ranchi. FILE PHOTO

districts,” Mr. Bauri said.

As per the schedule prepared, Mr. Sahu will be in Ranchi on September 8, Seraikela on September 9, and Khunti on September 11. Similarly, Leader of the Opposition Babulal Marandi will be in Dumka on September 8, Pakur on September 9, Sahibganj on September 10, and Godda on September 11.

Former Chief Minister Raghubar Das will be in Hazaribagh on September 8, Dhanbad on September 10, and Lohardaga on September 11.

Also, former Chief Minister Madhu Koda will be in Jamshedpur on September 8, Bokaro on September 9, and Ramgarh on September 11, whereas former national president of the ST Morcha, Samir

Oraon, will be in Simdega on September 9, Gumla and Chaibasa on September 10, and Giridih on September 11.

Union Minister of State for Defence Sanjay Seth will be in Garhwa on September 8, and former State president Ravindra Rai will be in Chatra on September 10 and Jamtara on September 11.

Ultimatum issued A BJP delegation had recently issued an ultimatum to the State government, conveyed through the Chief Secretary, demanding the withdrawal of cases registered against students, but the State government remained silent on the matter. Following this, the BJP decided on the Statewide agitation.

Vijay's broadside at DMK sparks uproar in Tamil Nadu Assembly

B. Kolappan
CHENNAI

The Tamil Nadu Assembly was in uproar on Monday after Chief Minister C. Joseph Vijay launched a scathing attack on the DMK and its leaders, accusing them of corruption and referring to the Sarkaria Commission report of the 1970s, the Veeranam “scandal”, and the alleged collection of “party funds” by what he called the “Karrur” gang.

Last week, Mr. Vijay had said he would make his reply and warned the DMK members not to “run away”.

Slogans raised DMK MLAs, who remained silent until 9.56 a.m. when the Chief Minister began his reply, started raising slogans first and displaying pamphlets later. They surrounded the



Chief Minister Vijay speaking in the Assembly on Monday. YOUTUBE/@TAMILNADUASSEMBLY

Speaker and even resorted to a sit-in after Mr. Vijay departed from his prepared speech, raised his voice, and targeted the DMK, case by case, accusing its leaders of nepotism.

He further cited songs of AIADMK founder and former Chief Minister M.G. Ramachandran (MGR) and the words of late Tamil poet Kannadasan to attack the DMK.

Cong. leaders on their way to Saharanpur detained for hours

Mayank Kumar
LUCKNOW

The Uttar Pradesh Police on Monday detained several Congress leaders, including State unit president Ajay Rai, in Lucknow while on their way to Saharanpur following the demolition of a century-old mosque in the Saharanpur Collectorate complex two days earlier following a court order. The Congress leaders were released hours later in the evening.

On Sunday, Mr. Rai had announced that a fact-finding delegation would visit Saharanpur to assess the situation.

Alleging that the administration is working on the directions of the Bharatiya Janata Party government to silence the Opposition, Mr. Rai said, “Following the bulldozing of the mosque, we were going to take stock of the situation in Saharanpur, but we were detained by the police as soon as we left the party office. This authoritarian action is clear evidence of the fact that the government is afraid of the Congress party's democratic pushback. This is a struggle against injustice and dictatorship.”

In Saharanpur, a Samajwadi Party delegation submitted a petition to the district authorities addressed to the President of India seeking action against officers responsible for the demolition. Opposition leaders earlier questioned the “hurried demolition” of the mosque just three days after a district court ruling.

Meanwhile, an Archaeological Survey of India team began examining the well found during the removal of debris on Sunday, triggering a debate over its historical significance. Congress MP Imran Masood said the well was part of the mosque and was used in earlier times to meet the requirement of water for wuzu (ablution) before prayers.

Baiga children deaths: panel questions delay in response

Sobhana K. Nair
NEW DELHI

Members of the Parliamentary Standing Committee on Social Justice and Empowerment questioned the government over its response to the deaths of Baiga tribe children in Madhya Pradesh's Balaghat district, asking why there were no clear answers on the cause of the deaths several months after the first cases were reported.

The panel met representatives of the Ministry of Tribal Affairs, the Ministry of Jal Shakti, and the Department of Telecommunications to review the implementation of the Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan.

77 days and counting: Haryana teachers seek job regularisation

Ashok Kumar
GURUGRAM

Guest teachers in Haryana's government schools have intensified their agitation seeking regularisation of services, with their protest in Kurukshetra entering its 77th day on Monday.

The teachers are demanding implementation of the State government's regularisation policies framed in 2014, citing recent judgments of the Supreme Court and the Punjab and Haryana High Court which, they say, have strengthened their claim for regularisation.

The teachers have been staging the protest outside the Secretariat in Kurukshetra, which is the home city of Chief Minister Nayab Singh Saini. Two teachers sit on 24-hour rotation every day, according to the Rajkiya Anubandhit Ad-

hyapak Sangh, which is spearheading the protest.

“We have won the case in the High Court as well as the Supreme Court. As far as I understand, the government has no legal option left,” Dinesh Yadav, State president of the association, told *The Hindu*.

The government, however, has not yet initiated the process of regularisation. Mr. Yadav said a delegation of teachers had met the Education Minister, the Additional Chief Secretary and other senior officials, but no concrete assurance has been given so far.

Bhupinder Singh, State general secretary of the association, said around 12,000 guest teachers are working on a contractual basis against sanctioned posts in government schools, and that nearly 55% of them are women.

The teachers, appointed



Guest teachers protesting outside the Secretariat in Kurukshetra, the home city of Haryana CM Nayab Singh Saini. SPECIAL ARRANGEMENT

between 2005 and 2008, are working as Junior Basic Teachers, Trained Graduate Teachers, and Post Graduate Teachers.

Two policies

At the centre of the dispute are two regularisation policies – a three-year policy and a 10-year policy – framed during the Congress regime in 2014. According to the association, the teachers had fulfilled

all conditions prescribed under the three-year policy but were not regularised. However, guest college teachers covered by the same policy were later given regular status, it said.

Mr. Singh claimed that the process of regularisation could not be completed as the model code of conduct came into force in the run-up to the Assembly election in 2014, which saw the BJP come to power for

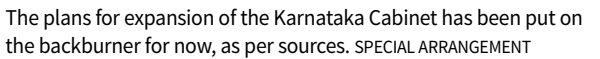
the first time in the State.

The teachers' case has gone through several rounds of litigation. The association said the Supreme Court, in its judgment dated April 16 this year, upheld the validity of the three-year policy while quashing the 10-year policy. The teachers subsequently approached the Punjab and Haryana High Court seeking implementation of the policy.

The association also cited the High Court's judgment dated May 27 as a key development. It said the court directed that eligible guest teachers be regularised under the 2014 policy within two months and criticised the prolonged continuation of teachers with an irregular status.

The teachers have since stepped up their agitation, including gheraoing the Chief Minister's residence





The Hindu Bureau
BENGALURU

"Any move to fill the va-

Meanwhile, another source said: "After the smooth and powerful launch of Mr. Shivakumar as Chief Minister, the developments around Cabinet expansion went off steam. The party is also worried that it could stir up the hornet's nest once again."

The Hindu Bureau
THIRUVANANTHAPURAM



However, the meeting took place at a time when Mr. Joseph had repeatedly expressed his desire to opt out of the key organisational responsibility. Several

The AICC had previously insisted that Congress

Suresh's claim

On Monday, Mr. Suresh told presspersons in New Delhi that being an MP or MLA did not preclude Congress workers from holding the post of PCC president. He pointed out that the AICC had nominated MPs and MLAs in Tamil Nadu, Assam, and Punjab recently.

The Hindu Bureau
HYDERABAD

Telangana Chief Minister A. Revanth Reddy has demanded that Leader of the Opposition K. Chandrababhanth Rao apologise to the institution of the Speaker and the people for Bharat Rashtra Samithi (BRS) MLC Tata Madhu's remarks, including the alleged objectionable language used against Speaker G. Prasad.

Mr. Reddy stated he had never heard such objectionable language against a Dalit leader and the institution of the Speaker anywhere.

Floor Leader of the Bharatiya Janata Party (BJP) A. Maheshwar Reddy condemned the BRS member's

Meanwhile, the BJP members staged a walkout after Payal Shankar demanded action against the BRS member's remarks against the Speaker. He requested the Speaker to check the genuineness of the video footage containing the member's reported remarks since AI-morphed and fake videos were damaging individuals.

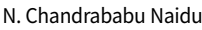
Veteran actor Mammootty turned 75 on Monday. Fans reached his residence in Kochi on Sunday night and distributed cakes even as the actor thanked them through a video call from Chennai. On his birthday, the actor took to social media to announce the first-look poster of his new movie *Mela*. In his greetings, Keralam Chief Minister V.D. Satheesan said “the pride of Malayalam” had created magic on the screen over the last five decades. Leader of the Opposition Pinarayi Vijayan said Mammootty has explored the immense possibilities of acting through his roles.

Close on the heels of the accident at Kaipamangalam in Thrissur that claimed the lives of eight students, Kerala Chief Minister V.D. Sathesnan has written to Union Minister for Road Transport and Highways Nitin Gadkari, seeking a comprehensive review of safety measures on the national highways in the State. The letter urged the Union Minister to direct the National Highways Authority of India (NHAI) to conduct an immediate safety review of all stretches of NH-66 where construction is in progress.

Holding mishandling by Air India Express crew of a special equipment booked by a customer as deficiency of service, a consumer court in Mangaluru has directed the airline to refund ₹2,000 that it had taken from the consumer as special equipment handling fee. It was told to pay a compensation of ₹20,000, and ₹5,000 towards litigation costs. The customer had found that his expensive equipment was placed on the baggage carousel along with other luggage. This was a contravention of the special equipment handling protocol for which he had paid a separate fee.

The Hindu Bureau
VIJAYAWADA

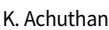
unleashed by the YSR Congress Party (YSRCP), he added. Addressing the TDP's Leadership Conclave at its central office near Mangalagiri, Mr. Naidu said the party would duly recog-



The YSRCP was running a false campaign and it should be exposed with concrete evidence, he said, citing the YSRCP's claims on the Veligonda and the Polavaram projects as an

The Hindu Bureau
PALAKKAD

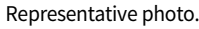
Mr. Achuthan represented the Chittur constituency in the Assembly for four consecutive terms from 1996 to 2016, winning elections in 1996, 2001, 2006 and 2011. He also served as president of the Thathamangalam Service Cooperative Bank and chairman of the Chittur-Thathamangalam Municipality. A farmer and cooperative activist,



He is survived by his wife C.K. Sudha, son Sumesh, who is the current MLA of Chittur, and daughter A. Sheeba, who is a lawyer. Leaders from different political and social circles condoled his death.

The Hindu Bureau
KOCHI

The application was developed by Eldho Vargh-



Grinson George, Director of the institute, said the application simplified biological data collection by automatically recording measurements and generating length-frequency datasets. The application allows multiple individuals

The Hindu Bureau
CHENNAI

The State government subsequently permitted a detailed inquiry, which was registered in December 2025.

The FIR was registered after permission was obtained under Section 17A of the Prevention of Corruption Act.



The Hindu Bureau
THIRUVANANTHAPURAM

He began his career as a journalist and later took to documentary filmmaking.

14 held as spurious liquor tragedy toll reaches 13 in M.P.

Mehul Malpani
BHOPAL

The official death toll in the hooch tragedy in Sagar district of Madhya Pradesh reached 13 after two more people died during treatment on Monday. State Congress president Jitu Patwari, however, alleged that the count was more than 50 after visiting affected villages.

The police, meanwhile, continued its crackdown and arrested 14 people linked to the network of spurious liquor, allegedly supplied from Lalitpur in Uttar Pradesh, officials said.

A magisterial inquiry led by a sub-divisional magistrate has been ordered into the matter to ascertain the circumstances that led to the deaths as well as the production and sale of the spurious liquor. The government has also announced a compensation of ₹4 lakh to the families of each of the deceased.

Earlier on the intervening night of September 5 and 6, several people from multiple villages in Sagar's Banda and Shahgarh subdivisions fell ill with vomit, fainting and seizures due to consumption of spurious liquor, officials have

The govt. has also announced a aid of ₹4 lakh to the families of each of the deceased

said. Deputy Chief Minister and Health Minister Rajendra Shukla, who has been deployed in Sagar, said on Monday that two more people died during treatment while more than 50 people are now out of danger.

In critical condition
“Three people are still critical and efforts are under way to save them. Ten people, small retail vendors who sell liquor in villages, were arrested [on Monday] and they are being interrogated. Very soon, we will reach the bottom of this and end it from the root,” he said. Sagar district PRO Soumya Samaiya told *The Hindu* that 92 people were still under treatment. “Medical teams are still deployed in the affected villages,” she said.

Earlier on Monday, the police booked 30 people in two separate cases under Bharatiya Nyaya Sanhita (BNS) Sections 105, 123 and 3(5), and provisions of the Madhya Pradesh Excise Act, 1915.

Manipur to update NRC with 1951 as base year, House told

Kuki-Zo Council urges the Centre to conduct the Census in the State; Home Minister informs the State Assembly that 14,992 out of 24,475 ‘illegal immigrants’ have been pushed back into Myanmar

The Hindu Bureau
GUWAHATI

Manipur Home Minister Konthoujam Govindas on Monday informed the 60-member Assembly that the National Register of Citizens (NRC) in the State would be updated using 1951 as the base year.

Manipur is divided on the NRC issue, for which the Centre recently deferred the Census exercise in the State. While Meitei and Naga organisations want the NRC before the Census and delimitation, Kuki organisations have called it “premature and unjustified”.

Mr. Govindas said that the State government would pursue the implementation of the NRC with the Centre in line with the resolutions adopted by the House. He cited certain figures to justify the resolutions (2022, 2024, and on September 2).

He told the House that 14,992 out of 24,475 “illegal immigrants” detected in the State so far have been pushed back into Myanmar. The immigrants entered the country through



Women hold placards during a sit-in protest demanding the conduct and update of NRC before the Census in Imphal. ANI

three border districts – Chandel, Kamjong, and Tengenoupal. Manipur has been grappling with the influx of undocumented Myanmar nationals since the military coup in the neighbouring country in February 2021. Insisting that the pushback process was being carried out through diplomatic channels, Mr. Govindas said the government instructed the Deputy Commissioners concerned to segregate the “illegal immigrants” from

local Indian communities.

The Home Minister further informed the House that the government constituted a 12-member panel on August 7, 2024, under the chairmanship of former Minister Awangbow Newmai to detect and identify the influx of illegal immigrants and the unnatural growth of new villages. A Cabinet Sub-Committee was also constituted on February 16, 2023, in this regard.

The Kuki-Zo Council

(KZC), the apex community body, on Monday urged the Centre to conduct the Census operation in the State, insisting that “politically motivated demands” should not obstruct it.

In a memorandum to Union Home Minister Amit Shah, the KZC said any decision concerning the NRC remained within the Centre's constitutional and legal authority.

The organisation also appealed to the Centre to re-impose President's Rule in Manipur if the State government continued to fail in its constitutional duty to protect all citizens, and expedite the political dialogue towards a “separate administration” in the form of a Union Territory with a legislature for the Kuki-Zo community.

“The Kuki-Zo people seek neither privilege nor preferential treatment. We seek equal protection, justice, security, and a political future in which our people can live with dignity and without fear,” the KZC said.

All Naga MLAs abstained from the final sitting of the Manipur Assembly session on Monday.

No ban on village volunteers, Arambai Tenggol: Manipur CM

The Hindu Bureau
GUWAHATI



Yumnam Khemchand Singh

Manipur Chief Minister Yumnam Khemchand Singh told the State Assembly on Monday that neither the “village volunteers” of the hills and the Imphal Valley, nor the Arambai Tenggol were banned organisations.

“Village volunteers” are armed civilian groups born out of the ethnic conflict that broke out between the Kuki-Zo and Meitei communities on May 3, 2023. The Imphal Valley-based Arambai Tenggol claims to be a socio-cultural revivalist group, but security analysts have characterised it as a radicalised armed militia.

The Chief Minister said the State government has been studying different cases of unlawful and criminal activities involving individuals during the conflict that left more than 300 people dead and displaced over 62,000 others.

‘Unprecedented crisis’
Taking part in a short-duration discussion moved by opposition legislator K. Ranjit Singh, the Chief Minister said that the unprecedented violence compelled many youth to take

up arms. “While some of them had good intentions, there could be others who are involved in criminal activities,” he said.

He said the police did not take up cases against such individuals based on their affiliation to the village volunteers or the Arambai Tenggol, as the State government has not banned them. “The arrests are based on individual-level criminal activities.”

He, however, said the State could not withdraw criminal cases filed against individuals during the crisis that broke out more than three years ago. “The cases against individuals will be studied one by one, and it will take time.”

Mr. Singh said the Manipur police were a very strong force in the past, but its morale plummeted following the crisis. “The government is trying to strengthen the force.”

Global watchdog says rise of ‘digital hawala’ is aiding money laundering, terror financing

Devesh K. Pandey
NEW DELHI

A centuries-old underground banking system is now transforming into a tech-enabled network, with virtual assets adding new layers to “digital hawala”.

The shift is being exploited for purposes ranging from routine money laundering to, in some cases, financing terrorist organisations, the latest Financial Action Task Force (FATF) report said.

The global money laundering and terror financing watchdog's report, titled “Investigating Professional Money Laundering, Underground Banking, and the Use of Hawala and Other Similar Service Providers”, identifies the fusion of virtual assets with the traditional “hawala” system as

one of the most significant developments. According to the report, nearly 70% of surveyed jurisdictions have reported the integration of new technologies into such networks.

Forms of digital hawala
The study has flagged six forms of “digital hawala” – a term covering a spectrum of technologies that facilitate the coordination, execution, settlement, or concealment of transactions.

One configuration involves digital coordination but traditional settlement, with operators using encrypted messaging applications, shared ledgers and online platforms to communicate instructions, recruit clients and couriers, and maintain records. The second is “digital customer interface”, such as mobile



The latest FATF report has flagged six forms of “digital hawala”. FILE PHOTO

wallets or fintech apps, although settlement between operators happens via cash or trade in both cases.

The next is virtual asset-based settlement, where stablecoins are used to settle balances directly. In the “integration with formal digital infrastructure” con-

figuration, funds are moved via payment service providers, fintech platforms, and virtual International Bank Account Numbers (IBANs).

The fifth is artificial intelligence-based tools for automated transaction structuring, dynamic mule-account routing, and high-speed fiat-to-cryptocurrency conversion. The last one involves “hawala” apps, which may be bundled digital ecosystems combining messaging, cloud storage, social media, Virtual Asset Service Providers (VASPs), lending apps, and even gaming platforms.

The FATF found that these configurations make transactions faster, more opaque and more complex, while expanding the network's reach and resilience.

The report said that digitisation is acting as a catalyst for traditional settlement mechanisms rather than replacing them, with cash remaining critical at the collection and exit points.

Turkiye case study

It cited a case study of Turkey's discovery of a “digital hawala” network financing the terror group, the Islamic State of Iraq and the Levant (ISIL). In 2023, raids on a self-described ISIL “administrative officer” led to the seizure of about \$57,250 in cash, followed by a further \$5,54,000, along with “hawala” notebooks and digital transfer receipts.

A jeweller and two mobile phone shops operated as fronts, and the transfers were disguised as charity donations.

Defence acquisition proposals worth ₹1.10 lakh crore cleared

Saurabh Trivedi
NEW DELHI

The Defence Acquisition Council (DAC), chaired by Defence Minister Rajnath Singh, on Monday accorded Acceptance of Necessity (AoN) for a range of acquisition proposals for the armed forces estimated at ₹1.10 lakh crore, with approximately 98% of the procurements planned to be from Indian industry.

According to the Defence Ministry, for the Army, the DAC approved the procurement of chemical, biological, radiological and nuclear (CBRN) reconnaissance vehicles, high mobility vehicles (HMTVs), self-propelled mechanical mine layers (MMLs), advanced light helicopters (ALHs), trawl tanks and the Sarvatra Bridge System.

The trawl tanks and the Sarvatra Bridge System will



Rajnath Singh

provide composite crossing capability to fighting formations during operations in varied terrains.

Domestic boost

For the Navy, the DAC granted AoN for the procurement of Arudhra radars and the design, development and subsequent procurement of marine gas turbines (MGTs).

The Arudhra radars will replace existing air route surveillance radars at va-

rious naval air stations. Indigenous development of MGTs, which are crucial for warship propulsion, is expected to reduce dependence on foreign vendors.

For the Air Force, the Council approved several proposals aimed at enhancing the war-fighting capabilities of its fighters, transport aircraft and helicopters.

The DAC also cleared the procurement of ground-based multi-purpose jammers (GBMPJ), which will provide effective jamming capability against radars. Approval was also accorded for the installation of the Defence Forces Secure Access Card (DEFSAC) system, which will replace existing paper-based identity cards, passes and permits with interoperable radio frequency identification (RFID)-based smart cards.

ED searches 7 places linked to Trinamool MP Nadimul Haque

Senjuti Sengupta
KOLKATA

The Enforcement Directorate on Monday morning began search operations across seven premises in Kolkata, Delhi and Ranchi as part of a money laundering investigation against Trinamool Congress Rajya Sabha MP Nadimul Haque, who is the proprietor of an Urdu daily.

Since 2004, the 56-year-old Mr. Haque, a journalist and businessman, has been serving as director of Akhbar-E-Mashriq Pvt Ltd, which publishes the daily *Akhbar-E-Mashriq*. Founded in 1980, the daily circulates its editions across six cities: Delhi, Kolkata, Ranchi, Lucknow, Bhopal, and Srinagar.

The MP has served as the head of the West Bengal Urdu Akademi.

According to the officials, the case pertains to allegations concerning the publication or dissemination of material promoting communal disharmony and also contains allegations regarding illegal transactions, unaccounted cash transactions, and funding from suspicious sources.

The case stemmed from an FIR lodged under the Bidhannagar Police Commissionerate.

Mr. Haque is also one of the Rajya Sabha MPs who have remained loyal to Mamata Banerjee after her defeat in the 2026 state assembly election.

Four Rajya Sabha MPs of the party have resigned, three of whom have later also been re-elected to the Upper House with the support of Bharatiya Janata Party.

Manipur musician killed in Delhi; 7 held, minor detained for lynching

The Hindu Bureau
NEW DELHI

A musician from Manipur, Chongtham Vikram Singh, living in Delhi's Ashram area, died on Monday of injuries suffered in an alleged assault by a group on Sunday night.

The Delhi Police, which registered an FIR under the Bharatiya Nyaya Sanhita (BNS) for mob lynching, arrested seven men and detained a 15-year-old boy in connection with the case.

The accused were engaged in packaging and delivering food from nearby dhabas and were aged between 15 and 36, the police said.

Prior threats

In his complaint to the police, Yaiphaba, the victim's son, said his father had told him on the way to the



Chongtham Vikram Singh

hospital that people from a nearby dhaba had attacked him. Some of them had previously threatened Singh after he complained of excessive noise, the complaint said.

Singh, a guitarist and musician from Imphal, has been part of rock bands but has not been working for the past few years. He was also a music teacher.

Former Manipur Chief Minister and BJP leader N.

Biren Singh said he was “deeply saddened” and “deeply concerned by the incident”. “With evidence now emerging in connection with the incident, I urge the Delhi Police to take swift and appropriate action, thoroughly investigate the matter, and ensure that justice is delivered at the earliest,” he said.

FIR registered

The Delhi Police registered an FIR against unknown persons under Sections 103(2) (involves murder on the grounds of race, caste or community by a group of five or more persons) and 3(5) (crimes committed by a group with common intent) of BNS.

The police said that “the circumstances leading to his assault and death are being ascertained and necessary legal action is being taken as per law”. The

police have identified the arrested as Bharat Kumar, Pramod, Ramzan Khan, Deepanshu, Mohan Vishwakarma, Vijay, and Mannu.

Around 11.30 p.m. on Sunday, Yaiphaba and his mother were at home when they heard loud noises and Singh screaming. Yaiphaba said that when he opened the door, he found four to five people assaulting his father, “kicking and punching him”. When he raised an alarm and tried to stop them, the attackers fled. Yaiphaba said he had seen the faces of some attackers.

Yaiphaba and his mother took Singh to a private hospital where he died on Monday morning. His family members said the body was taken to the All Institute of Medical Sciences for a post-mortem examination.

‘Hindutva influencer’ sent to 14-day judicial custody in assault case

The Hindu Bureau
NEW DELHI

A Delhi court on Monday remanded Hindutva influencer Swatantra Bhardwaj to 14 days in judicial custody in connection with the alleged assault of the father of a student during a protest at Jantar Mantar in July organised by the Cockroach Janta Party (CJP).

Additional Sessions Judge Saurabh Pratap Singh Laler of the Patiala House Courts passed the order after Mr. Bhardwaj was produced before the court following the completion of his one-day police custody. The court had remanded him to police custody for a day on Saturday. He will be produced before the court again on September 21. Mr. Bhardwaj was detained in Uttar Pradesh on Friday, a few

hours after the CJP held a demonstration outside the Parliament Street police station demanding his arrest in connection with the alleged assault.

The Delhi police subsequently added provisions of the SC/ST (Prevention of Atrocities) Act and criminal intimidation to the FIR registered against Mr. Bhardwaj over the alleged assault of the minor activist's father. A separate case under the POCSO Act has also been registered against him.

Mr. Bhardwaj came under scrutiny after a video podcast in which he claimed to have “cracked the skull” of the student's father during the demonstration went viral. He also claimed during the interview that he had political connections that helped him avoid arrest.

Rahul seeks accountability for Delhi building collapse

"Statistical jugglery cannot hide the impact of inflation. The public is watching everything," he said.

13 lakh voters deleted from Delhi's electoral rolls in pre-SIR phase

Rahul Gandhi interviews candidates for AICC posts

India's carbon credit scheme receives U.K. recognition: official

Gujarat to host National Film Awards for first time

Quota will put more women in political leadership: Om Birla

He also urged women legislators to share successful initiatives and strengthen awareness among women, especially those in vulnerable sections, about laws and institutional mechanisms available to address their concerns.

Congress SC, ST MPs to petition President over 'purification' row

Monsoon Session has been kept on 'ventilator': Congress

We allow Gen Z to express their opinions freely: Bengal BJP chief

His comments came after some Hindu outfit leaders called upon Bengali Hindus to avoid non-vegetarian food during Navratri.

A IN-Y

India can cut steel emissions before coal plants lock them in

India's green steel journey is still nascent but recent developments, policies have indicated that this hard-to-abate sunrise industry is rising up to the net-zero challenge; balancing growing demand with climate goals remains formidable, however, with a lot riding on the availability of green hydrogen

Neelima Vallangi

A slew of announcements over the last few months from the Indian government have placed steel sector decarbonisation high on the agenda of the country's steelmakers, including a ₹5,000-crore scheme announced last year to accelerate steel sector decarbonisation in the country that is set to launch in the coming months.

India's steelmaking emissions are among the highest in the world, roughly 32% higher than the global average, and account for nearly 12% of the country's total greenhouse gas emissions. Driven by strong demand from infrastructure development, construction, and automotive manufacturing, India's steel production in FY 2025-26 was the second-highest in the world, at around 160 million tonnes, behind only China.

However, steel is a notoriously hard-to-abate industry, and India has its own set of challenges in bringing steelmaking emissions down to zero.

Carbon-intensive method

The government's ongoing National Mission on Green Steel aims to lower steelmaking emissions intensity from the current 2.55-2.65 down to 2.2 tonnes of carbon dioxide-equivalent (tCO₂e) per tonne of crude steel by 2029-30. A certification scheme it launched under the Mission in 2024 said any steel manufactured with an emissions intensity of less than 2.2 tCO₂e will be considered "green" and the greenest steel could still have an intensity of up to 1.6 tCO₂e. Globally, the average emissions from steelmaking hover around 1.85 tCO₂e, making India's greenest steel still highly carbon-intensive.

"India is a developing economy, and we do not have much scrap available. If we have old steel available, we can recycle that scrap to produce new steel, which reduces our costs and emissions associated with reducing iron ore," Anubha Aggarwal, an analyst at the Centre for Research on Energy and Clean Air, an independent research organisation, said.

The traditional way to make steel involves burning coal to provide intense heat in a blast furnace, where iron ore is smelted into liquid form using coking coal to strip oxygen. The carbon-rich liquid is then treated with pure oxygen at high speed in a basic oxygen furnace to convert it into high-quality steel by burning away all the impurities. This method, called BF-BOF (blast furnace-basic oxygen furnace), is widely used worldwide, and it is highly carbon-intensive.

Reducing emissions to produce green steel will involve skipping the first step entirely, when possible, by reusing steel scrap to produce new steel and replacing coal-powered blast furnaces with electric arc furnaces (EAFs), which use electricity to produce steel from scrap, or direct reduced iron (DRI). Steel with a near-zero carbon footprint (excluding scope-3 emissions) could also use green hydrogen instead of natural gas to reduce iron ore and be powered by clean electricity generated from renewable sources.

However, 70.4% of global steelmaking still uses BF-BOF, according to a *Nature Reviews Clean Technology* analysis. The remaining uses the EAF method, which has less than half the emissions intensity of BF-BOF steel.

'Just a redirection'

In India, the steel industry is relatively more heterogeneous: as per an India Steel Association report, 43% of total crude steel is produced at BF-BOF plants, 22% from EAF units, and 35% from electric induction furnaces (which use electromagnetic fields to process small batches of scrap or DRI and are known to be energy-efficient). Given the growing demand for steel and several planned steel plants, BF-BOF capacity is projected to increase to 56% by 2030.

In these circumstances, India essentially has a large window of opportunity to become one of the biggest players in green steel and iron production – if India decides to go the extra mile and invest in greener pathways rather than in coal-based steel, Clara Bachorz, a researcher working on steel sector decarbonisation at the Potsdam Institute for Climate Impact Research, Germany, and lead author of a recent paper investigating pathways to avert future emissions from steel industry through strategic green investments, said.

"And this [greener pathway] also has high potential for emission reductions compared to the cost of reducing those



The traditional way to make steel involves burning coal to provide intense heat in a blast furnace, where iron ore is smelted into liquid form using coking coal to strip oxygen. Representative image. YASIN HEMMATI/UNSPASH

emissions later, mostly using bioenergy with carbon capture and storage or carbon dioxide removals," she added.

According to the paper, published in *Nature Climate Change*, avoiding new BF-BOFs and not relining young furnaces that are due for one, and diverting the investments to building EAFs, could almost halve global committed steel emissions.

"We find that since the majority of these BOF plants in India have not broken ground, you could redirect investments already towards DRI-EAF, and this would not require additional investments. It would just be a redirection," Dr. Bachorz said.

India's steel plants are indeed headed for relining blast furnaces, a capital-intensive but essential maintenance task done every two decades that can extend a plant's lifetime by another 15-20 years – but while locking in coal-based steel production and high carbon emissions for that time. More than 43 million tonnes per annum of blast furnace capacity is due for relining before 2030, according to an analysis by Reclimatize, an independent research body examining India's decarbonisation efforts in hard-to-abate sectors.

Switching to hydrogen

In their study, Bachorz et al. modelled two global steel sector transition pathways aligned with 1.5 degree celsius of warming by the end of the century, with overshoot of up to 1.7 degree celsius. They found China and India dominate the dynamics of the fast-transition scenario, driven by their near-term BF-BOF investments that far exceed those of other regions.

In the slow-transition scenario, young blast furnaces are relined and announced BF-BOF plants are developed as planned, locking in emissions for decades and relying significantly more on currently nascent carbon dioxide removal

technologies in future, in addition to deep emission cuts in other sectors to compensate for steel sector emissions.

In effect, the team's analysis showed early investment shifts – particularly in rapidly growing economies such as India – can substantially reduce future emissions at comparably low costs, making early action in steel a high-leverage climate mitigation opportunity, as in the fast transition scenario.

"Today, it is the most mature green steelmaking technology that you operate, either with natural gas, which still has emissions of around 1.2 tCO₂ per tonne of steel or with green hydrogen, which is then almost zero emissions," Dr. Bachorz said.

According to the team's model, India first operates these DRI-EAF plants primarily with natural gas until 2040-2045, after which it switches to hydrogen as green hydrogen becomes sufficiently cost-competitive. "If you wanted to operate such a plant with green hydrogen today, my feeling is that this would be a lot more expensive," she continued.

In conclusion, the researchers suggested steel is less a 'hard-to-abate' sector than a sector facing a 'hard-to-abate barrier' that can be overcome at relatively moderate costs through timely investment decisions.

Defining 'green steel'

"When you compare a sector like steel with aviation, it is relatively much easier to abate because the technology is already there, even though it is more expensive," Dr. Bachorz said.

"So you need to pay the green premium and basically go over this investment barrier. But once you commit, it is one of the easiest hard-to-abate sectors."

"If we invest in EAF now, it would be much easier for us to make it cleaner later

rather than investing in BF-BOF now and it becoming a stranded property later," Ms. Aggarwal said. "Because we have already seen this with thermal power plants, where it becomes really difficult to close them down, as a lot of investment has gone into them, and you want a return on your investments. We do not want to repeat the same story with steel."

However, she warned that natural gas-based EAFs also carry the risks of fuel shortages, import dependence, geopolitical shocks, and stranded natural gas infrastructure (when used as a bridge fuel to reduce steel emissions).

ArcelorMittal Nippon Steel India became the first integrated steel producer to receive green steel certification in February this year. According to government data, 89 steel units had been awarded the same certification as of March 31, covering a production volume of 12.34 million tonnes.

"You may have your doubts about whether [India's 'green steel'] is really green, and the brackets for green steel certification are too large not to be very effective, but then at the end of the day, we have taken that initiative of defining what 'green steel' is, unlike anywhere else in the world," Ms. Aggarwal added. "What we have not done is create the kind of market where there is also demand for green steel."

'Future-proof pathway'

This January, the European Union's Carbon Border Adjustment Mechanism came into effect, imposing steep penalties on carbon-intensive imports, including India's high-emission steel, which is exported to several European countries. In response, India has been seeking alternative markets and boosting domestic consumption, while the market and compliance obligations may be forcing the steel industry to rethink investment decisions and adopt steel production technologies with lower emissions intensity.

In March, the Union Ministry of New and Renewable Energy funded three pilot green hydrogen projects totalling more than Rs. 400 crore to pursue innovation and validate the technical feasibility of using 100% green hydrogen in furnaces and other stages for low-carbon steel production. Last year, JSW Energy commissioned its first, and India's largest, commercial-scale green hydrogen plant in Karnataka's Vijayanagar to supply 100% green hydrogen to adjoining JSW Steel's DRI unit.

ArcelorMittal Nippon Steel India, in the meantime, is promoting its green steel to buyers interested in slashing their scope-3 emissions.

India's green steel journey is still nascent but recent developments and policies have indicated that this hard-to-abate sunrise industry is rising up to the net-zero challenge. However, balancing growing demand with climate goals remains a formidable challenge for now, with a lot riding on green hydrogen availability and affordability.

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John Ternus (left) joined Apple's cinema display team in 2001 after a stint designing VR headsets at a small firm called Virtual Research Systems. GETTY IMAGES VIA AFP

How John Ternus's era will shape Apple

John Xavier

Under Tim Cook, Apple attained \$4.6 trillion market capitalisation, built an efficient supply chain network that sold over three billion iPhones, and effectively stewarded Steve Jobs' legacy. While Mr. Cook's era will be remembered for scale and operational efficiency, those twin markers of growth could not have been accomplished without an underlying solid product line-up. That is where hardware engineering comes in – the unit headed by John Ternus.

Mr. Ternus joined Apple's cinema display team in 2001 after a stint designing VR headsets at a small firm called Virtual Research Systems. His first project at Apple was the Cinema Display monitor. The company had just launched the iPod and was basking in its glory. But consumers were shifting to cellular devices as the world was coming online.

Apple needed a product to capture the trend as Finnish handset maker Nokia's 3310 mobile phones – shipped with small, green-lit display screens – were selling like hot cakes. While these devices were solid, their functionality was limited. Yet they filled the gap by increasing storage space that could be used by buyers to store files that included music. This was eating into the music player market.

Ternus has helped shape some of Apple's most important products, like the iPad and AirPods, while championing innovations such as iPadOS to open full potential of Apple's hardware

With the help of French computer scientist Jean-Marie Hullot, who wrote programmes for the original Macintosh and operating system for a smartphone, Apple built the iPhone, a device that redefined mobile telephony by ditching physical keypad – a design change that elbowed BlackBerry aside in the high-end mobile market.

In 2010, Apple unveiled iPad, a device that further solidified its position in making an industry-leading consumer hardware. In 2013, the same year it launched iPad Air, the company elevated Mr. Ternus as Vice-President of hardware engineering, a move that recognised his contributions by enhancing his span and control over product lines.

By 2021, he was made the global head of hardware engineering.

Apple's own succession language credits him with being instrumental in introducing multiple new product lines. Mr. Ternus has also been known for pushing for the creation of iPadOS back in 2019 as a distinct operating system rather than leaving the iPad running a stripped-down version of iOS.

But the clearest preview of the Ternus era is the iPhone Air. It gave a glimpse into how Mr. Ternus builds and operates. Rather than simply shrinking an existing iPhone, his team coordinated closely with hardware technologies team led by Johny Srouji to fit pro-level performance into a dramatically thinner chassis.

That meant a new A19 Pro chip, Apple's N1 wireless chip, and – most notably – the C1X modem, Apple's second-generation in-house cellular modem, which matters enormously for a phone with a thinner, smaller battery.

The iPhone Air shows how Mr. Ternus's hardware engineering teams and Mr. Srouji's chip teams work in lockstep from the earliest stages of a product's design. The modem, the processor, and the physical chassis were engineered together rather than in sequence.

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Natural partners

India and Belgium are building ties across sectors despite differences

In terms of size and engagement, Belgium is not one of India's most important partners. However, the visit of the Belgian Prime Minister to Delhi is part of a larger puzzle of India's ties with European countries. These have now gained salience with focus areas for India: in economic relations, defence relations and strategic ties. At the top of the agenda in talks between Prime Ministers Bart De Wever and Narendra Modi was bilateral trade opportunities once the India-European Union Free Trade Agreement (FTA) is signed – which is expected in December. India-Belgium trade is about \$13 billion, a third of which is from over a century of collaboration in the diamond industry. Antwerp is a global hub for diamond trading, while Mumbai and Surat provide the biggest cutting and polishing services. However, once the EU FTA is signed, they hope to benefit from collaboration in other areas including renewable energy and food processing; the Prime Ministers agreed to set a target of doubling bilateral trade in the next five years. Despite being surrounded by friendly European allies, Belgium has a robust high-tech defence industry – small arms, drone and counter-drone technology, ammunition and naval defence – and is keen to make inroads in the Indian market, with India seeking to diversify its procurement beyond global powers. The two sides agreed on an MoU for co-development of military hardware, and signed a letter of intent on defence cooperation. India will appoint a resident defence attaché to Brussels, complementing Belgium's move.

Statements by the two leaders, however, indicated that the visit went beyond the bilateral to a more geopolitical framing of strategic ties, both bilateral as well as between India and Europe. At a business event, Mr. De Wever said that Europe is now realising the truth of India's "warnings" on strategic autonomy from global powers and over dependence in trade and industrial production, in particular referring to China. After the talks, Mr. Modi said that democratic values, a market economy, and people-to-people ties make India and Belgium "natural partners", a theme he has promoted this past year, as he has stepped up visits to Europe and received a number of European leaders in Delhi, in the run-up to his Brussels visit later this year for the FTA. It is also significant that Mr. De Wever's visit comes in the same week that New Delhi is preparing to welcome Russian President Vladimir Putin for the BRICS summit in Delhi. The engagement with Belgium demonstrates a new maturity for both European countries and India. This shows sensitivity to their respective positions on the Russia-Ukraine conflict, indicating that ties outweigh Europe's enmity and India's deep friendship with Moscow.

Ground control

ISRO must focus on space technology as a tool for development

The irony was evident: on the day the Indian Space Research Organisation (ISRO) had its greatest success of the year – the launch of its first geosynchronous imaging satellite (EOS-05) on a temperamental GSLV – its employee organisations demanded clarity on its long-term plans. But this grievance was long in the making. Since the Indian Space Policy of April 2023, the message has been clear that ISRO would, one day in the unspecified future, stop making commercial satellites and launch rockets. Clearly inspired by NASA, it expects to focus on ambitious exploratory missions and, literally, moonshots. Why would a space agency ramping up space missions upset a section of employees? This is not the paradox it presents. The letter, co-signed by nine employee associations, and addressed to the chairman, seeks clarity on staff-strength, recruitment and the outsourcing of core functions. ISRO says that it will "not be privatised or reduced"; IN-SPaCE, set up to facilitate private participation, says the agency will "not be diminished", only that the industry's role must grow. Both are silent on the core concern – potential job losses. This brings up an underlying tension that ISRO must deal with: does its vision of India's space sector align with its founding principles, of not being enamoured of space races and being laser-focused on space technology as a tool for social development? Or is the image of power, where India is one of a handful of space-faring countries with a mushrooming of 'space startups', now the bedrock of that vision?

With a budget 16 times the Department of Space's – \$24.4 billion against ₹13,705 crore – NASA too has scaled back since the 1960s: from 0.7% of American GDP in 1966 to 0.1% now, and from 36,000 civil servants at Apollo's peak to about 14,000 today. Unlike NASA, which in its early years designed and made every nut and bolt, ISRO has always had a manufacturing relationship with the private sector (Walchandnagar, L&T). What is different is that the buzz is not about established companies building on experience but new entrants with fleeting foreign capital, most interested in satellite data as a service. No sector can hope to be static; survival can mean significant changes to how it operates. There is yet no serious reckoning with the might of China's space programme. If India's space sector in 2035 is to be known as a source of export earnings, a nucleus of value-added services that brings in jobs and absorbs skilled labour, then hard choices taken today may be worth it. But jumping onto a bandwagon of space – as fuel for the science fiction fantasy of billionaires and colonial notions of conquest, to align with some misplaced notion of an 'exclusive' club – is foolhardy. ISRO must have a transparent policy that spells this out.

BRICS has strayed quite far from where it started. To be held this weekend (September 12-13, 2026) the 18th BRICS Summit, in New Delhi, gives India a huge opportunity to bring coherence back to the expanded group and give it direction if it is to be an effective body in shaping global events.

BRICS was formed to give a greater voice to Brazil, Russia, India, China and South Africa, the emerging economies, in global governance and institutions, particularly financial and economic institutions, and to aim for a more equitable multilateral order. India was initially an enthusiastic participant, seeing the group as a vehicle for genuine reform of multilateral institutions. The first summit hosted by India (the fourth BRICS summit), in 2012, focused on the theme "Global Stability, Security and Prosperity," tacitly reflecting a much broader geopolitical canvas. The Indian presidency helped lead to the establishment of the BRICS Bank, the New Development Bank. Other Indian initiatives followed, including the integration of a counter-terrorism architecture into BRICS' work.

The emergence of the China challenge

However, China had other plans. At the time, even if it was not anti-West, China saw BRICS as a counterpoise to western domination – a view that India also initially shared in the context of reform. More importantly, China needed a group such as BRICS, consisting of emerging economies, to amplify its global ambitions, given that BRICS countries accounted for nearly 20% of world GDP in 2010. Further, China wanted to use BRICS mechanisms such as BRICS-Plus and BRICS Outreach to reach out to the Global South and consolidate their profile as a second pole in a future bipolar world.

Suddenly, India not only had a reform agenda on its hands but also had to act as a counterweight to China's larger designs on BRICS. India, Brazil and South Africa soon realised that Chinese support for the reform of multilateral institutions was selective. For example, China resisted supporting the three countries' bid for permanent seats on the United Nations Security Council. Consequently, India started focusing more on BRICS consolidation and intra-BRICS matters to resist Chinese grandstanding. In fact, this year's theme for India's chairship has four pillars – Resilience, Innovation, Cooperation and Sustainability – and draws on Prime Minister Narendra Modi's vision of "Humanity First" and a "people-centric" approach to BRICS cooperation.

In addition, there was a real danger that the principle of consensus, which had kept BRICS a cohesive group, would be bypassed and that BRICS would go the way of the Shanghai Cooperation Organisation (SCO), where the majority can bulldoze their views through. India and Brazil resisted these efforts. The 2019 BRICS Summit in Brasilia was held without any other invited guest countries – just the five original members. But in 2020, under the Russian presidency, probably the first non-consensus chair's statement on COVID-19 was issued,



T.S. Tirumurti

Chair, Steering Committee, Deccan Centre for International Relations, former Permanent Representative of India to the United Nations, New York, and former Sherpa of BRICS

The Delhi summit offers India an opportunity to restore BRICS's original purpose

bypassing the sacred principle of consensus. If non-consensus documents are increasingly resorted to, as we saw at the BRICS Foreign Ministers' meeting in May 2026 in New Delhi due to the inability of the new members to agree, then BRICS will cease to be effective.

Membership expansion tests BRICS cohesion

But then, it was difficult to resist the Chinese juggernaut. China pushed for the expansion of the New Development Bank and then for the expansion of BRICS itself. India resisted, as did Brazil. But that was not to be, and BRICS now has 11 members, following the addition of Egypt, Ethiopia, Iran, Saudi Arabia, the United Arab Emirates and Indonesia, as well as 10 partner countries.

India's sagacity has now been vindicated, as the new members of BRICS have started fighting among themselves. While the original five have their differences, some of them serious, they know how to address them with maturity for the greater good of the group. The new members do not have any such compunctions and are subverting BRICS through their bilateral conflicts. BRICS should not go the way of the South Asian Association for Regional Cooperation (SAARC), with bilateral disputes paralysing the group.

In addition, the tussle over the direction BRICS should take has begun. India has tried to keep BRICS as a "non-West" group, as opposed to the increasing pressure from some members to turn BRICS into an anti-West group. An anti-West slant is becoming increasingly difficult to avoid, with China competing with the United States for global leadership, Russia at war with Ukraine with the full backing of Europe, and Iran being bombarded by the U.S. and Israel. It becomes even harder to maintain the "non-West" line when Brazil and India have been subjected to punitive tariffs by U.S. President Donald Trump, and the U.S. Congress is considering legislation that would give the President the power to levy punitive tariffs on countries importing Russian oil. The question being asked is: how can India stop BRICS from becoming anti-West at a time when the Quad (Australia, India, Japan, the U.S.) is being emasculated by the U.S., India-U.S. relations are under great pressure, Pakistan is being courted by the U.S. at India's expense, and global institutions are being made dysfunctional by the West? The more erratic the U.S. is *vis-à-vis* BRICS, the greater the chances of BRICS going in the wrong direction. Fortunately, India's effort to prevent an anti-western drift is shared by many other BRICS members, since they too have much to gain by engaging with multiple actors from different camps in their national interest. They want change, but not geopolitical realignment with China and Russia. This is the direction of BRICS's future.

But there is a much larger question. Both the U.S. and China are enunciating parallel visions of the world and playing by their own rules, rather than internationally negotiated ones. Both are, in effect, challenging existing international norms and global institutions and setting up parallel

structures and standards in emerging areas such as Artificial Intelligence, digital and Internet governance, data ownership, state control, 5G/6G and telecom, satellite navigation, and electric vehicles. Neither is looking for reform of existing institutions. China has just set up a new World AI Cooperation Organisation in Shanghai, in addition to other parallel initiatives such as the Asian Infrastructure Investment Bank, the Belt and Road Initiative and the Digital Silk Road, which challenge the Bretton Woods institutions and their governance and financing models. They see BRICS as one of the vehicles through which to pursue their broader goals. For example, the push for de-dollarisation through the establishment of a parallel BRICS currency is receiving a lukewarm response, since many members are uncomfortable with a renminbi-dominated currency and would prefer merely interlinking payment systems, central bank digital currencies and national-currency transactions. Nevertheless, after the success of its pilot project, China is formally launching its ambitious mBridge, an alternative financial payment system. The last thing India wants is an alternative Bretton Woods system dominated by China.s

India can course correct BRICS

The 2026 Summit in Delhi provides an opportunity for course correction, and India has worked hard to persuade the new members to overcome their political differences. In a world in flux, India has to revive the original *raison d'être* of BRICS and shape the expanded grouping into a vehicle for reform and stability, while stopping others from hijacking it.

At the 2018 BRICS Summit in South Africa, Mr. Modi first articulated his vision of "reformed multilateralism" at the leaders' retreat. By 2019, this vision had found its way into the BRICS Summit document in Brasilia, with India, Brazil and South Africa pushing for it. The time has come to revive this agenda and make BRICS a strong voice for the "non-West" middle powers – and, by extension, for the Global South.

In Davos, the Canadian Prime Minister spoke about broadly West-centric middle powers coming together. A Global South middle power is very different. And if there is a credible organisation of Global South middle powers, it is BRICS. It has its problems. To begin with, China is hardly a middle power and has disproportionate influence within BRICS. India and China need to find greater synergy on emerging global issues, even as their bilateral differences are being contained. Further, some middle powers that ought to be there are not represented in BRICS. However imperfect its composition may be, it is clear that BRICS has great geopolitical and geo-economic potential.

At a time when the profile of BRICS is only increasing, with many countries wanting to join it, India should proactively embrace the BRICS reform agenda to serve its larger interest of multi-alignment. If India does not, others will take BRICS in a different direction.

U.S. tariffs are not what is holding back Indian research

Tariff wars with the United States have ebbed and flowed for years. After months of escalation, Washington and New Delhi stepped back a few months ago. The penalty tied to India's purchases of Russian oil went, the reciprocal tariff was cut, and medicines and most electronics stayed exempt. Exporters treated it as relief. But an old worry in policy circles is worth testing: that high American tariffs will choke research in the industries that they touch, and set back India's move into higher-value production.

The reality

That worry rests on a mistaken assumption that the sectors exposed to American tariffs are the sectors where India does its research.

Consider the impact of the 2019 loss of duty-free access and the 2025 tariff hit on sectors such as organic chemicals, plastics, base metals, machinery, auto components, and leather. Now consider how much these industries do in research. Indian metals firms spend about 0.4% of sales on R&D, compared with a global average of nearly 1.6%. Auto and parts makers spend a little over 2%, versus a global average of 5%, while electrical equipment firms spend less than 2%, against the same 5% global benchmark. These industries were research-thin long before any tariff arrived. The sectors most exposed to trade shocks and the sectors that invest in research barely overlap. India's research effort is concentrated in pharmaceuticals and automobiles, and almost everything else does little. A tariff on chemicals or steel cannot cut research spending that was never there.

This also takes the air out of the claim that tariffs are holding back Indian innovation. Patenting and research spending in the exposed sectors show no clear break during the tariff years that can be attributed to the duties. These sectors had been on a low-research path well



Arindam Goswami

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The sectors most exposed to trade shocks barely overlap with those that invest in research

before 2018 and stayed on it. The tariffs are a symptom of a trading relationship under strain.

Research is a fixed and risky investment that pays off only when a product sells at scale. A firm facing a smaller export market can retreat to cheaper, undifferentiated goods, or invest in products that a tariff cannot easily replace. Indian industry has chosen the first path for decades, and a tariff does not change that habit.

Where tariffs do matter is narrow. Of the two sectors that account for most of India's research, pharmaceuticals secured an exemption in the February deal, while automobiles did not. The 25% U.S. duty on auto parts remains, and metal tariffs raise input costs for downstream engineering and component firms. So, the live risk to India's research base runs through autos – not the broad manufacturing economy that fills the headlines.

The real problem

The problem to worry about predates this trade fight and will outlast it. India spends a very small percentage of its output on research, a figure that remains well below that of the countries it wants to compete with. The private industry share is also well below the global leaders. What industry does spend tends to go toward routine development and testing rather than research that creates new products. By one comparison, Nvidia alone spends nearly as much on research as all of Indian industry combined.

This is the backdrop for the government's ₹1 lakh crore Research, Development and Innovation scheme, launched late last year. It offers long-tenure, low-cost capital to firms in sunrise and strategic areas such as artificial intelligence, semiconductors, quantum technology and biotechnology. The ambition is overdue. But the money is pointed at the frontier, while the industries most exposed to trade shocks

and most in need of moving up the value chain are older sectors that the scheme is not designed for. A chemicals or auto-parts maker that has never run a research programme will not start now because of a loan aimed at deep tech.

That is the opportunity the tariff episode offers. A trade shock tells an exposed industry that its products are vulnerable. The useful response is to move those firms into differentiated products that a tariff cannot easily undercut, which means investing in research. Incentives can be weighted towards core research rather than routine testing and aimed at exposed sectors instead of being spread thin. The higher costs that metal tariffs cause on downstream engineering firms can be offset for those that keep up research spending. Pharmaceuticals and autos also deserve protection in any further negotiations, because tariffs in these sectors can affect innovation. Sheltering an exposed industry without asking anything in return preserves low-value production. Support should be tied to research effort, so that relief rebuilds capability rather than freezing the status quo.

Time to measure research better

None of this works without better measurement. India's official research figures arrive years late and undercount private spending. A country that cannot see where its research is going cannot steer it. A faster, firm-level record linking research spending to exports would let policymakers act before the damage shows.

The tariff truce gives India time. The tariffs were never the main thing holding back Indian research. The task is to fix what is: an economy whose exposed industries do too little research, while its research strength is in a few sectors that could be vulnerable to the next trade shock.

The views expressed are personal

LETTERS TO THE EDITOR

A dam, the concerns

China's proposed colossal dam on the Yarlung Tsangpo has precipitated profound geopolitical and ecological apprehensions. The region's pronounced seismicity renders such an enormous hydraulic structure particularly

disquieting. The absence of an enforceable transboundary water-sharing mechanism further exacerbates India's predicament. China must institutionalise hydrological transparency. **Nagarajamani M.V.,** Hyderabad

Assembly proceedings

Of late, the Tamil Nadu Assembly appears to have become more of a shooting spot than a place for serious debate. Members shout at the top of their voices, while 'cheerleaders' from the respective political parties

enthusiastically thump the tables — often without even understanding what is actually being discussed. There was much hype surrounding the Chief Minister's response to the points raised by the Leader of the Opposition. But in the end, it turned out to be

nothing more than a damp squib. Allegations were made without producing evidence, while the much-talked-about 'files' are still waiting to be opened. Only a handful of members appear to be focusing on people's welfare and the real issues affecting

them. Meanwhile, the larger concern remains the deterioration of law and order in Tamil Nadu.

R. Shankar, Chennai

Letters emailed to letters@thehindu.co.in must carry the postal address and the full name.

Teen agency beyond Meta’s reckoning

Meryl is 13. She describes YouTube Shorts as “so addicting”; holding her attention “just long enough” that she loses track of everything else. She is one of dozens of teens who have participated in the research at the Teens, Family, and Technology Lab at Rutgers University. While the 13-year-old’s language is not unusual, teen voices are still largely missing from broader debates about social media use.

Meta’s \$17.1 billion settlement with U.S. States is making headlines as a reckoning over addictive platform design. However, not much has been said about how teen agency emerges in a media environment built to predict the most tempting feed. Debates about youth and technology have historically focused on media effects: either what teens do with media, or what media does to teens. Nearly 100 years ago, the Payne Fund Studies marked the first systematic exploration of this dynamic, examining the impact of movies on children. With every new wave of technology, the storyline persists. Youth audiences are typically spoken for – cast as passive, trapped or targeted by new tech.

This framing misses the point. It ignores how the experience of adolescence itself shapes media use, and why certain types of media design captivate teens. Teens’ media use resonates with a powerful framework from developmental psychology: a race car that can go very fast, but whose brakes are still being built. That speed creates both opportunities and risks. Teens are primed to learn



Nikhila Natarajan

Co-founder of the Teens, Family, and Technology Lab at Rutgers University

more rapidly than at any other stage of development. Yet that power makes teens more vulnerable to negative stimuli. This heightened sensitivity exists alongside a crucial asymmetry: their ability to plan and prioritise (what to do first, what next) develops gradually, even as teens exhibit heightened sensitivity to “rewards” (easily persuaded by things that are pleasing). In today’s media, these rewards arrive via personalised recommendations. When short-form video loops shrink the gap between media experiences to 15 or 30 seconds, the onus of decision-making becomes the young person’s problem at a time when the braking system is still getting built.

Whistleblowers, Silicon Valley defectors, and documentaries such as *The Social Dilemma* have laid bare how platform design choices erode human agency. What remains absent from public commentary is what teens’ vocabularies reveal about their developmental needs. Across the 13-to-16 age group, teens say that short-form videos are shrinking their attention spans. But their reflections point to a persistent entry point: boredom.

Boredom, peer dynamics
Boredom is a developmental driver. As adolescents shift their orientation toward peers and crave greater independence, they continue to live under adult restrictions. They then open an app seeking a quick distraction, only to find themselves stuck for hours on platforms such as YouTube Shorts or TikTok.

Peer dynamics drive how teens move across different apps, with platform design acting as an amplifier. On platforms such as Instagram and Snapchat, where popularity is quantified

through public likes and follower counts, every interaction gets a scoreboard. For an adolescent hyper-sensitive to social inclusion and exclusion, this design is incredibly stressful. Early evidence from Australia’s social media ban bears this out. Two systematic surveys by Bursztyn and colleagues found that Australian teens report needing roughly 70% of their peers to quit social media before they’re willing to quit themselves.

Research at the Teens, Family, and Technology Lab shows that adolescents navigate an average of 7 to 12 different platforms daily. Instagram and Facebook exist within a competitive teen media inventory alongside Snapchat, YouTube, Pinterest, Discord, Spotify, and more. Here, teens’ recalibration of media practices, and parents’ concerns for their children’s wellbeing act as powerful competitive forces.

While Meta and other platforms chase topline revenues, some teens are crafting micro-frictions to push back. Pinterest is a big hit with girls who want to escape performative sociality; music streaming competes with short-form video apps; and some teens are deleting apps intermittently. These are behavioural signatures of adolescent metacognition – literally, thinking about thinking. Yet AI-fuelled media design continues to operate at the limits of adolescent development, exacerbating tensions between the intentional and accidental. Understanding the contradictions of adolescent development invites us to reconsider teen media use in a new light – as a striving for agency.

The incongruity of caste and science

All humans must be treated equally. That is a moral and constitutional position. It is not a scientific position because science does not carry moral or constitutional authority.

The normative system laid out in the ‘Manusmriti’ expresses the ‘principle’ that the birth of a person says something essential about who that person is supposed to be, with persons born into specific social categories to be assigned a place in the division of labour and a set of permitted relationships and social rights. Modern constitutional thinking is radically different. Per the Magna Carta, the English Bill of Rights and the constitutional democracies of today, individuals possess rights that depend on more than their place in a social hierarchy. After all, the fundamental rights of the Indian Constitution are attached to persons and citizens, not to inherited social functions.

Yet a poster that turned up at IIT-Mandi on September 4 said the role of “Brahmanas” is to “spread god’s message”, that of “kshatriyas” to “protect society and spirituality”, of “vaishyas” to “drive economy, support others”, and of “shudras” to “serve higher classes”.

Science, which underlies the technologies the IITs are primarily concerned with, offers another conception paralleling the constitutional view, rooted in a different kind of authority.

The authority of science
While some defenders of the caste system have claimed that it is based on inherited biological differences, the system is really the cause of the differences, rather than their inheritor. The human



Vasudevan Mukunth

organism develops as its genes interact with its environment, and caste-based discrimination affects these environments. Studies on the social determinants of health have revealed caste-related inequities in healthcare and access to healthcare while social epigenetics has found that social circumstances can induce biological changes via mechanisms that affect the way genes are regulated.

People of Black African heritage and Indigenous peoples in Asia, say, have non-identical genomes for this reason. By applying science’s methods, we know the differences between these peoples do not excuse a caste-like system. Instead, the differences in these people’s social circumstances through history – including societies of which they were part practising a hereditary, hierarchical system at any point – affected the way they developed.

In a landmark 2009 study, geneticists reported that most present-day Indians descended from two differentiated ancient populations, which mixed extensively before endogamy took root, and present Indian groups contain mixes of these ancestries. The team also said there are substantial genetic differences between various communities today thanks to a long history of endogamy. Other studies of Indian populations have also reported strong founder effects (when a new population emerges from a small group of individuals and inherits the lack of genetic diversity) due to endogamy. The genome does not announce that these people were ordained

to marry one another but records the fact that they elected to do so for reasons beyond scientific wisdom.

The human nervous system is also very plastic: it constantly rewires itself in response to every experience in a life-long process. Not everyone is good at everything, but plasticity means the neurological system develops together with each human’s environment, interests, temperament, and individual choice, giving rise to a variety of abilities among people.

No matter how well-equipped, the practice of science could never say whether all humans are equal. That is a political and ethical proposition it cannot test or prove. However, it does say that all humans are members of one species, that their development is shaped by their experiences, and that social environments can become embodied.

Just as caste constrains the plasticity of the human brain, it also constrains the human capacity for culture. *Homo sapiens* can accumulate knowledge and pass it on so that other *H. sapiens* do not have to learn from scratch. Caste-based discrimination, however, restricts the spread of knowledge and narrows the social range over which one of the defining capacities of our species can operate – as if a student’s ability to solve a differential equation has anything to do with the status of their ancestors.

This clash of possibilities is what makes the appearance of a message reinforcing the caste system at a centre of scientific learning particularly tragic.

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Nearly 8,000 people died due to collapse of structures in 5 years

While the incident in Satya Niketan in Delhi has sparked outrage, hundreds of people have been dying due to laxities in enforcing building safety norms.

DATA POINT

Pon Vasanth B.A

The collapse of a five-storey building in Satya Niketan in Delhi on September 6, in which six people have been confirmed dead as of Monday morning, has turned out to be yet another grim reminder of the laxity in cracking down on buildings violating safety norms.

While collapses like the one in Satya Niketan have caught the public’s attention, because of the high death toll and the fact that it was a PG for students, many such isolated incidents which happen across the country go largely unnoticed.

An analysis of the annual Accidental Deaths and Suicides in India (ADSI) reports showed that a total of 7,874 people were killed in building collapses across the country between 2020 and 2024. Delhi recorded 169 deaths in the same period.

The ADSI 2024 report – the latest available from the National Crime Records Bureau – stated that while such deaths showed a marginal decline in the country in 2024, the deaths in the National Capital Territory (NCT) have increased since 2020 (**Chart 1**).

The ADSI includes five sub-categories under deaths due to collapse of structures – Collapse of Dwelling House/Residential Building, Collapse of Official/Commercial Building, Collapse of Dam, Collapse of Bridge, and Others. The data discussed in this article excludes the numbers reported under the categories of ‘Collapse of Dam’ and ‘Collapse of Bridge’.

Uttar Pradesh, the most populous State in the country, recorded 223 deaths due to building collapses in 2024, the highest in the country. It was followed by the second most populous State, Maharashtra (as per Census 2011) with 220 deaths. **Chart 2** shows the 15 States and UTs that reported the

highest number of deaths due to building collapses.

Delhi reported the 12th highest number of deaths (42) in the country in 2024. Despite being geographically smaller, the densely-populated NCT has been among the top 15 States every year since 2020. Moreover, the deaths recurring every year in the capital are more worrying when seen in relation to its population. Deaths from building collapses per million population, calculated using mid-year population estimates from the ADSI reports every year, show that Delhi recorded the second- or third-highest deaths per million population in 2022, 2023, and 2024 (**Chart 3**).

Recent numbers

While the ADSI report for 2025 is yet to be released, a compilation of incidents reported in the media showed that at least 70 people have died in 20 incidents since January 2025, which means that at least three people have died every month in such incidents.

Map 1 shows the incidents reported in the media since 2024, including the one in Satya Niketan on Sunday. There have been at least 34 incidents, killing more than 90 people since 2024.

Acting in predictable haste, as most government agencies do after such incidents, the Municipal Corporation of Delhi, has said that it will seal all illegal five-storey and taller buildings “immediately”.

However, data discussed in this article indicates that the underlying systemic problem demands well-thought-out long-term strategies that keep in mind the growing need for affordable housing to end such avoidable deaths.

Note on map: The incidents and death toll were compiled from news reports. Though each incident has been verified from at least three news reports, there may be some inaccuracies in the numbers. Incident locations are geocoded to locality level, and not the precise spot of the incident.

Accountability collapse

The data for the charts were sourced from annual Accidental Deaths and Suicides in India reports. DataMeet’s Delhi municipal spatial data was used to plot the map. Incidents plotted in the map were sourced from media reports. With inputs from Gauri Singavarapu and Gandla Sneha, who are interning with The Hindu’s Data team



CHART 1: Deaths due to building collapses have increased in Delhi. Y-axis scales are different for All India and Delhi

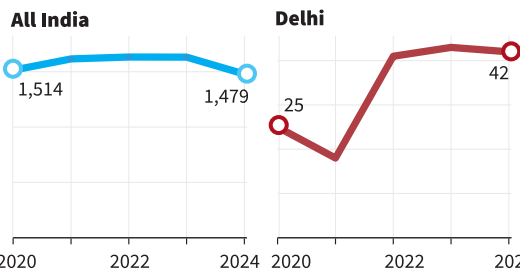
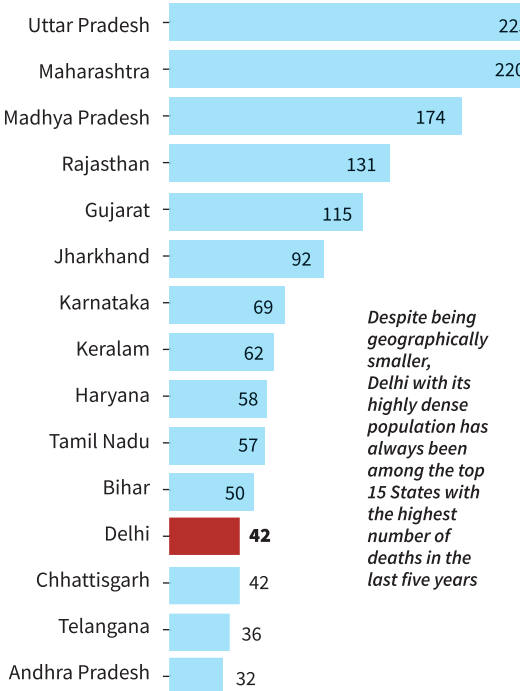
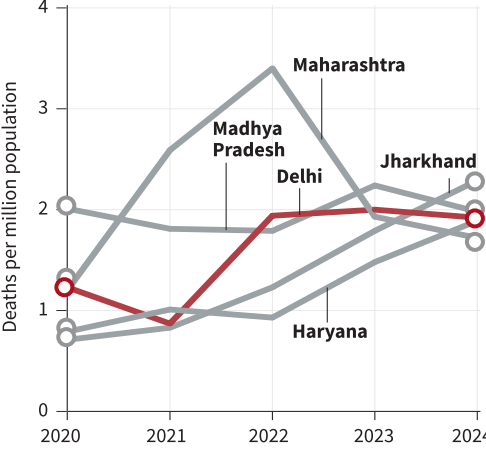


CHART 2: The 15 States (or UTs) with the highest number of deaths due to building collapses in 2024

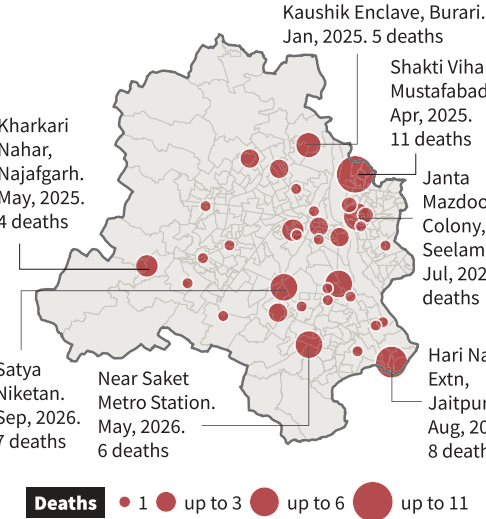


Despite being geographically smaller, Delhi with its highly dense population has always been among the top 15 States with the highest number of deaths in the last five years

CHART 3: Delhi recorded the second or third highest deaths per million due to building collapses since 2022



MAP 1: In at least 34 incidents of building collapses since 2024 in various parts of Delhi, over 90 people were killed



FROM THE ARCHIVES

The Hindu

FIFTY YEARS AGO SEPTEMBER 8, 1976

Bhutto lays claim to Kohinoor diamond

London, September 7: Pakistan has asked Britain to return the fabulous Kohinoor diamond which is the central stone in the Queen Consort’s crown and has been in British possession since 1849.

A Government spokesman said to-day the request was made in a personal letter to the Prime Minister, Mr. James Callaghan by the Pakistani Premier, Z. A. Bhutto. It was “being considered”.

The 100 carat gem now stored in the Tower of London is acknowledged as the finest diamond in the world and became part of the crown jewels of Queen Victoria after the British annexed Punjab. The 1,000-year-old diamond was last worn by the Queen Mother at the Coronation of King George VI in 1937.

A spokesman for Queen Elizabeth to-day would not comment on the monarch’s authority to accede to President Bhutto’s request.

The palace spokesman said Prime Minister Callaghan was likely to discuss the request with the Queen.

Informed sources said the Pakistani Premier wrote in his letter that the return of the stone would be a demonstration of the spirit that moved Britain to shed its imperial encumbrances.

The request for the coveted Kohinoor – ‘Mountain of Light’ – might become another chapter in the turbulent history of the stone.

A HUNDRED YEARS AGO SEPTEMBER 8, 1926

South Africa’s status

Capetown, September 7: General Hertzog’s speech at a meeting in the City Hall on the eve of his departure to attend the Imperial Conference dealing with the Flag Bill has foreshadowed a compromise and he declared that if the Committee on the flag should recommend the inclusion of a crown in it he was confident that the Government would approve it.

Gen. Hertzog denied that he intended at the conference to demand a written constitution and said he intended to urge that steps should be taken so that South Africa’s national status which was equal to that of Britain and other dominions be entitled to an international recognition and should be so published and declared to the world.

THE HINDU

2,799 The police in Kerala registered cases against 2,799 people for allegedly driving under the influence of alcohol during a week-long special drive across the State, a police spokesman said on Monday. PTI

COMPILED BY THE HINDU DATA TEAM

CACHE

Can air-cooling handle the heat from a 1-GW data centre?

Air-cooling uses less water and is cheaper and well-established, but it cannot practically handle the heat load of a 1-GW data centre on its own; while liquid-cooling imposes higher costs upfront, air-cooling imposes a performance and efficiency tax

Vasudevan Mukunth

The story so far:

Following sustained local protests and pressure from civil society groups, Google has reportedly said its planned 1-GW data centre in Visakhapatnam district will use air-cooling technology to cool its servers. While air-cooling requires less water, it comes with its own trade-offs.

Why does a large data centre generate heat?

A large data centre will have several million processors, each with billions of transistors. Each transistor manipulates the electrical current flowing through it to perform operations on data. As it does, the resistance the transistor's material poses to the flow of current will release some heat, as will the charging and discharging of transistors. Across trillions of transistors, the amount of heat thus generated can be considerable.

A 1-GW data centre will in principle produce 1 GW of heat. The planned Google 'hyperscaler' near Visakhapatnam and the newly announced Tata Consultancy Services (TCS) 'HyperVault' will each be 1-GW facilities. How much of that power is for the data centre alone is unclear. Assuming it is 1 GW: each facility will effectively have to remove 1 GW of heat out of and into the environment. In the very first step, the heat from each chip is moved from the silicon die holding it to a designated heat sink, such as a cooling plate. But what happens from the heat sink depends on which cooling technology the facility uses.

How does a data centre move heat?

Since the heat comes from transistors, it helps to understand the architecture first. Billions of transistors make a processor. One or more processors make up a server. Multiple servers fill a rack. Multiple racks are combined into clusters. Multiple clusters make up one data centre.

There are several cooling options depending on economic feasibility, scale of operations, and local conditions.

In air-cooling, fans push cool air through or around the server racks, and the hot air at the end is collected and cooled. Some common configurations include computer-room ACs (CRACs), which use ACs with refrigerants; computer-room air-handlers (CRAHs), which use chilled water to cool the air; hot-aisle/cold-aisle containment, where hot and cool air are kept physically separate; and free-cooling, where heat from servers is simply expelled into naturally cooler outside air. A related version is the rear-door heat exchanger: air leaving the servers passes through a heat-exchanger attached to the back of each rack, where chilled water absorbs the heat, and the cooler air is reflowed to the servers.

Another popular option is direct liquid cooling, where a specific liquid is brought in contact with the components generating heat. In one version called cold-plate cooling, a metal plate embedded with channels to carry the liquid is placed directly against a processor. The liquid draws the heat via the plate and flows from there via tubes to a heat-exchanger, where it loses the heat, before returning. Liquid-cooling is usually more effective than air-cooling because water-based liquids typically have higher



Close encounters: An Amazon Web Services data centre adjacent to single-family homes in Stone Ridge, Virginia, U.S. AFP

heat capacity than air, for the same volume. The heat capacity of a substance is the amount of heat it has to be given to raise its temperature by, say, 1° C. So the higher its heat capacity, the more heat it can absorb before itself warming up.

That said, the server can still air-cool those components that do not merit liquid-cooling. Third is an option called immersion cooling, where the electronics themselves are immersed in a non-conductive liquid. In single-phase immersion, the liquid draws heat, which is pumped away through a heat-exchanger. In two-phase immersion, the liquid boils as it draws heat; the heat is drawn away, e.g., by a fan, and the vapour condenses back. The latter is useful because heat transfer is more efficient at higher temperatures.

Because the heat-sink of a liquid is in contact with the electronics, the heat is moved more efficiently, and so immersion cooling can handle very high power densities. However, it requires specialised hardware and maintaining the coolant through complex processes.

Fourth is evaporative cooling: the heat is removed from the servers, e.g., using air-cooling, then transferred to water, which is then allowed to evaporate in cooling towers or evaporative condensers. If the local air isn't very humid, evaporative cooling can be highly energy-efficient. However, it requires large quantities of water.

Another similar option is dry-cooling, where a heat-exchanger is simply exposed to the ambient air and fitted with large fins to encourage the heat to leave. While this avoids the water demand of evaporative cooling, it also incurs larger heat-exchange surfaces, especially in places with warm weather.

Then there are chilled-water systems that use a central chilled water plant; geothermal heat rejection, an experimental technology where the heat is pumped into heat-exchangers underground; natural water cooling, where a large, natural water body like a nearby lake is used as the heat sink; and

heat reuse, where the heat is moved to serve applications like industrial processes or desalination.

All of them come with trade-offs. E.g. the geothermal option is hard to scale up while reusing heat is hard without heat pumps because the heat usually needs to be at a higher temperature to be useful.

Is air-cooling good?

Generally, air-cooling imposes lower infrastructure costs up front than a liquid-cooling setup. Current construction benchmarks suggest liquid-cooled facilities will incur a 7-10% premium on the total project cost due to the demand for complex piping and leak detection systems and special coolant distribution units.

Air-cooling is also a proven technology by now and comes with a deep and mature pool of technicians and maintenance protocols. In fact, irrespective of the cooling technology used in a cluster's heat hotspots, engineers often prefer to air-cool racks producing 20 kW or less of heat. It is an added bonus if the ambient conditions are naturally cool or arid, in which case air-side economisers that pull in outside air for cooling purposes can reduce the energy diverted to mechanical cooling by 70%. But all this said, as of 2026, using air-cooling alone to cool a 1-GW data centre is considered impractical, both physically and economically. While liquid-cooling imposes higher costs upfront, air-cooling levies a performance and efficiency tax, so to speak, that can increase the total cost of ownership for an air-cooled 1-GW facility within a few years of operation.

The most important reason is the 'thermal wall': air-cooling can remove at best around 40 kW per rack. However, modern artificial intelligence (AI) chips like the Nvidia Blackwell generate 700-1,000 W per GPU, so a rack could emit around 120-150 kW. At this scale, air-cooling the racks will require them to be installed inside a wind tunnel, with the power consumption skyrocketing.

Air-cooling will also demand acoustic control as the number of fans and chillers required will create large amounts of noise.

In fact, air-cooling chillers and air-handling units can produce up to 100 dB, and if the resulting low-frequency hum is not quickly dampened, it can travel several kilometres if atmospheric conditions are also favourable, disturbing the sleep of people in the way. For this mix of pros and cons, many 1-GW data centres today use hybrid cooling: air-cooling for low-density servers and other areas, rear-door heat-exchangers to push air-cooling to remove up to 50 kW, and direct-to-chip (DTC) liquid-cooling for high-density AI clusters, including the Nvidia Blackwell.

What have Google and TCS committed to?

After concerns were raised about the water demands of the Google hyperscaler data centre, Visakhapatnam MP M. Sriharat said, "Google has confirmed it will use air-cooling technology... there is no question of impact to water supply."

The tech giant has also said it will replenish 120% of water consumed for non-cooling needs across its operations by 2030, quantified under the Volumetric Water Benefit Accounting standard. *Hindustan Times* also quoted Alexander Smith, principal of Global Infrastructure and Energy, Asia Pacific, Google, as saying "Google is also planning integrated watershed-management measures, including rainwater catchment systems on data-centre buildings, groundwater recharge infrastructure, water pumps, and water ATMs".

Likewise, *Mint* quoted HyperVault CEO Deepesh Kiran Nanda saying, "We are building infrastructure for where AI is going: higher density, liquid cooling, larger power blocks, and faster deployment". Per news agency ANI, the HyperVault will also feature DTC cooling as well as use renewable energy.

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THE DAILY QUIZ

Following the 'Correct the Map' resolution adopted by the UN last week, a quiz on maps and geography

V.V. Ramanan

QUESTION 1

Which African country sponsored the resolution for the 'Equal Earth projection', and which was the lone nation to vote against the resolution, as against 164 supporting the same?

QUESTION 2

What is the scientific study of maps called and which entity is the principal agency in India responsible for all national map publications?

QUESTION 3

What is the name given to the network of latitude and longitude lines drawn on a map or globe?

QUESTION 4

The Prime Meridian passes through five African countries. Two of them are Togo and Ghana. Name the other three.

QUESTION 5

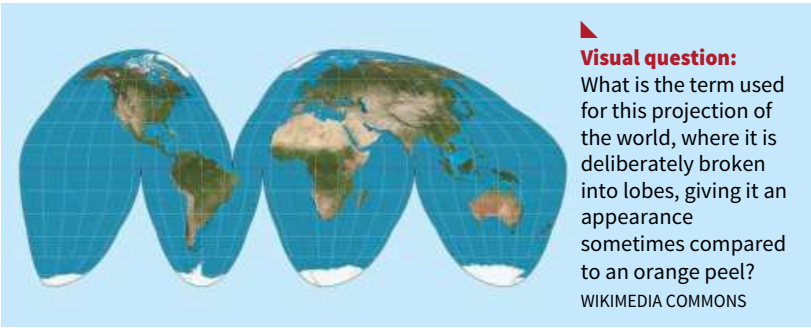
What remote point in the South Pacific, formally called the oceanic pole of inaccessibility, is farther from land than any other location in the world's oceans?

QUESTION 6

Map-makers sometimes use 'trap streets' (or 'fake towns') while designing maps. What are they?

QUESTION 7

Why is the term 'atlas' used for a book containing maps?



Questions and Answers to the previous day's daily quiz:

1. Childhood - Public speaker - Pune ashram - Arrival in the United States - 1984 bioterror attack. **Ans: Osho (Rajneesh)**
2. Early life - Mathematics career - Life in Montana - Mail bombings - Manifesto. **Ans: Ted Kaczynski, aka the Unabomber**

3. Early life - Founding a blood-testing startup - Self-made female billionaire - Downfall - Fraud charges. **Ans: Elizabeth Holmes, founder of Theranos**
4. Early life - Career - Photography hobby - Later years - Death - Discovery and recognition - Legacy. **Ans: Vivian Maier, American street photographer**
5. Early religious and political influences -

Founding a new religious organisation - Relocation to South American country - Mass murder-suicide. **Ans: Jim Jones**

6. Early life - Led her own gang - Massacre - Surrender - Political career - Assassination. **Ans: Phoolan Devi**
Visual: Identify the African leader and fill in the blank. Early life - Rise in Army - Seizure of power - Expulsion of _____ - War with Tanzania - Exile - Death - Erratic behaviour, self-bestowed titles and media portrayal. **Ans: Idi Amin; Ugandan Asians**
Early Birds: Sekar S. | A.C. Vineeth Kumar | Piyali Tuli | Aryan Wari | Siddhartha Viswanathan



FROM THE ARCHIVES

Know your English

S. Upendran

What is the meaning of "fly by night" operators? (M.K.B. Nambiar, Mahe)

Every month we read in the papers that the manager of some chit fund company has disappeared with the hard-earned money of his clients.

Such people can be called fly by night operators. They set up a company, do roaring business for about six months and then they suddenly disappear. They are there one day and then perform the disappearing act overnight.

The word "fly" in the expression means to "flee", that is, to run away. Since the shady characters who run such businesses leave town during the dead of night when everyone else is sound asleep, they are called "fly by night operators". All this build-up should have given you a fairly good idea as to what the expression means.

It means "undependable, dishonest". The expression can be used with people as well as with organisations.

There are many fly by night characters staying in that hotel.

Did you know that you could refer to a person as "fly"? When you say that someone is "fly", it means he/she is very knowledgeable about something. This use of the word "fly" is considered slang. So do not use it in formal contexts.

Amrit is fly, there's no doubt about it.
The word also means "stylish, nice-looking".

Some of the girls think that the present Vice Chancellor is really fly.

What is the collective noun for tigers? (G. V. Chalapathi Rao, Nellore)

Tigers are solitary animals; therefore, there is no specific collective noun to refer to them. Here are some interesting ones, however: pride of lions, school of fish, pod of whales, gaggle of geese, and rag of colts.

How is the word "pandemonium" pronounced? (M. Venkatesan, Srivaikuntam)

This is a word that many people in our country have begun to associate with Parliament sessions. The first syllable is pronounced like the word "pan", while the "e" in the second syllable and the "u" in the final syllable are pronounced like the "a" in "China". The "mon" rhymes with "bone", "phone", and "cone". The "i" in the fourth syllable is like the "i" in "it", "bit", and "fit". The main stress is on the third syllable, "mon". Any idea what the word means? When there is pandemonium, there is usually a lot of shouting and screaming because people are angry or frightened about something.

What is the difference between "leave" and "holiday"? (P. Maheswari, Venkatagiri)

A "leave" is something that you take on your own; it is something that you apply for and when it is granted, you are allowed to be away from work for some time.

The word "holiday" comes from "holy day"; originally, it was a day of rest from work for everyone. Nowadays, there's nothing holy about a holiday. If you say that tomorrow is a holiday, what you are saying is that everyone, not just you, is getting the day off. Unlike "leave" one needn't apply for "holiday".
Published in The Hindu on November 20, 2001

Word of the day

Alacrity:

liveliness and eagerness

Synonyms: willingness, enthusiasm, goodwill, swiftness

Usage: *The team welcomed the news with alacrity.*

Pronunciation: newsth.live/alacrity

International Phonetic

Alphabet: /əˈlækrəti/

For feedback and suggestions for Text & Context, please write to **letters@thehindu.co.in** with the subject 'Text & Context'

EU announces funding boost for Greenland as Trump seeks control



Israel threatens 'full-scale war' on PA in case of October 7-style attack



Nepal holds a day of mourning as rescuers focus on trapped workers



Thousands turn out in Serbia for funeral of convicted war criminal



U.S. energy facilities in Gulf remain exposed to retaliation, warns Iran

Far-right AfD wins key German State poll, falls short of majority

U.K. govt. slams anti-migrant protests

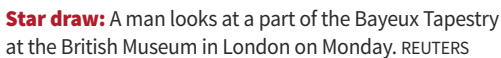
King in the rubble



Houthis accuse Saudi Arabia after missile attack on prison in northern Yemen kills seven

North Korea adds second destroyer as Kim urges stronger nuclear deterrent

Russia and North Korea already shared rail and air links.



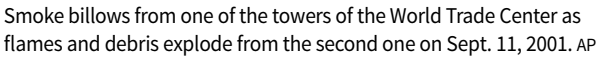
Associated Press
LONDON

Governments on both sides of the Channel have hailed its loan as a symbol of a Franco-British friendship. After decades of failed British requests to borrow the tapestry, an agreement was announced in 2018, delayed by COVID-19 and Brexit tensions, and finally confirmed by Mr. Macron during a state visit to the U.K. last year.

Associated Press
NEW YORK

It's a tapestry of mean-

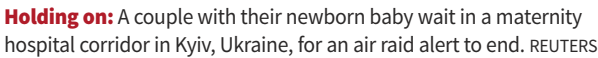
Different networks overlap in Jenna McPartland's



“My middle boy wrote his college entrance essay on being the child of a 9/11 family, but not being the 9/11 kid. And how he wouldn’t exist had his

Immigration had already been a contentious issue before the attack: Who should be allowed to come? Who could stay? What did it mean to be American, and who could claim it? After September 11, it took on another dimension of added fear and suspicion. And much of it fell on people of Muslim faith or coming from Mus-

Reuters
KYIV



The same was true for

In an escalating air war between Ukraine and Russia, in which each side is pounding the other's energy and logistical assets to force an advantage, Moscow has fired swarms of high-speed jet drones that are more destructive than

Beneath the hospital, several pregnant women walked into a shelter with their partners in response to a fresh air raid warning.

"We wanted a boy. But there's a war going on, and I don't need a boy. Let's just have girls."

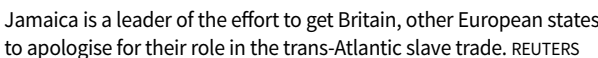
The cooling act: A woman walks past a water fountain in front of the main entrance of the Alte Oper (Old Opera) in Frankfurt am Main, western Germany, on Monday, as temperatures reached 31 degrees Celsius in the region. AFP

Associated Press
LONDON

While the petition is formally addressed to the King, he has no personal say on whether it will be referred to the Judicial Committee of the Privy Council, and he will act on the ad-

Publicity about the Zong massacre helped fuel the campaign to end slavery in the British Empire. Britain abolished the slave trade in 1807, but it wasn't until August 1, 1834, that the enslavement of human beings was outlawed.

Although he has stopped short of endorsing reparations, the King has repeatedly expressed his personal commitment to



Buckingham Palace said Monday that it has been working closely with the Governor-General of Ja-

When the United Nations General Assembly in

contrary to English common law? Were the forced

Jamaica's petition is based on an 1833 law that made the Judicial Committee of the Privy Council the highest court for civil and criminal appeals in the British Empire. When the

But rejection of the petition could still be a win for Jamaica.

LEARNING CURVE

The captaincy process leads to immense personal growth: Kishan



When you are the captain, the first thing is situational awareness, you have to stay alert the entire day on the field to understand what is happening and what isn't. Going through this whole process leads to immense personal growth, and while you might not realise it immediately, your mindset shifts completely — Ishan Kishan.

TEAM FIRST

‘When you are the skipper, it becomes a duty to help players’



There are many small things we often overlook in our daily routines. You don't look at your players or try to help them figure out how they can get better. But when you are the skipper, it becomes your duty to help them, because if your teammates are improving, ultimately you will achieve the team's goals — Ishan Kishan.

SMOOTH START

Maaya storms into second round of junior singles at New York



Maaya Rajeshwaran outclassed Emery Combs of USA 6-2, 6-2 in the first round of the girls singles at the US Open. The 15th-seeded Indian will take on Arina Malygina in the second round. In the doubles event, Maaya and Antonina Sushkova will meet the American duo of Combs and Olivia De Los Reyes in the first round.

‘SPECIAL MENTOR’

Pietersen joins England coaching team ahead of World Cup



Kevin Pietersen has joined England's white-ball coaching team as a “specialist mentor” in the build-up to next year's World Cup in southern Africa. The former England captain will be part of Brendon McCullum's coaching group for the upcoming T20 and ODI series at home to Sri Lanka later this month, as well as subsequent series.



A thorn in the flesh: Kishan made the South bowlers toil for a second successive day. R. RAGU

Kishan puts South under the pump

The southpaw's blistering double century takes East past the 700-run mark; Kushagra joins the party with a hundred against a hapless attack

DULEEP TROPHY FINAL

S. Dipak Ragav
CHENNAI

Ishan Kishan has been on a rollercoaster ride over the last few years. From being part of all three formats in 2023, he hit rock bottom by early 2024 when he was dropped from the BCCI central contract list, partly for his refusal to turn up for the Ranji Trophy. Since then, the Jharkhand wicketkeeper-batter has been making amends, putting in the hard yards in domestic cricket to forge his way back into India's T20 and ODI squads. Now, the 28-year-old is showing that he is ready to do the same in the longest format as well. He hammered 270 (334b, 30x4, 7x6) for East Zone against South Zone on day two of the Duleep Trophy final at the MAC Stadium on Monday. Resuming at 385 for four, East posted a gargantuan total of 708 with Kumar Kushagra (101, 132b,

8x4, 5x6) completing his century. Kishan and Kushagra added 230 for the fifth wicket, pummelling a hapless South attack into submission. Kishan set the tone with a brace of boundaries off Mohammed Siraj and M.D. Nidheesh. Kushagra soon joined in the act, lofting C.V. Milind and driving Nidheesh down the ground. He raced through his 90s by taking on Shreyas Gopal and brought up his hundred. South's only hope of staging a comeback was with the second new ball, but Siraj, who bowled just three overs in his first spell, was off-colour yet again. Post lunch, Kishan tore into Shreyas, smashing the leggie for three boundaries in an over. What stood out in the captain's knock was the way he shunned risks even when he used his feet, often drilling the ball past the bowler all along the ground. He raised his second First Class double century with a searing cover drive off Devdutt Padikkal. After the milestone,

Kishan switched to T20 mode and unleashed a barrage of sixes, lofting spinners Shreyas and T. Vijay with disdain. He needed just 30 more deliveries to raise his bat again – for a 250 – despite battling cramps. Barring a miracle over the last three days, Kishan has ensured East has one hand on its third Duleep Trophy title. The scores: East Zone — 1st innings: Abhimanyu Easwaran run out 72, Vaibhav Sooryavanshi c sub (Tanay) b Milind 44, Shikhar Mohan b Milind 85, Sudip Kumar Gharani c Milind b Vijay 6, Ishan Kishan c Smaran b Vijay 270, Kumar Kushagra c Milind b Vijay 101, Anukul Roy lbw b Rasheed 45, Md. Kounain Quraishi c Siraj b Shreyas 40, Mohammed Shami b Shreyas 28, Abhijit Sarkar (not out) 0, Mukesh Kumar c & b Vijay 0; Extras (b-8, lb-2, nb-6, w-1): 17; Total (in 163 overs): 708. Fall of wickets: 1-100, 2-133, 3-147, 4-241, 5-471, 6-626, 7-680, 8-698, 9-708. South Zone bowling: Siraj 25-3-140-0, Nidheesh 15-2-58-0, Vijay 42-4-180-4, Milind 21-2-88-2, Shreyas 42-7-165-2, Rasheed 13-0-46-1, Padikkal 4-0-18-0, Bhui 1-0-3-0.

‘Captaincy has helped me become a better batter’

One of the hardest tasks for an international cricketer is how they respond to being dropped from the national squad and fight their way back. And over the last year, Ishan Kishan has shown the way to do it. “Sometimes, when we play regularly for the Indian team, you just keep playing in a flow. You may be a very good player, which is why you're playing for India, but your growth stagnates at a certain point. “When you take one step back and go back to



Kishan... in full flow. R. RAGU

here on Monday. “You start thinking, ‘I need to get even better,’ because today, there are so many players scoring runs. To get back into the team, your only option is to perform much better than them. “You get time to reflect on what you missed earlier, why you couldn't convert your starts into big scores. I mean, the batting doesn't change that much; it's just the mindset that changes over time.” Last year, Kishan led Jharkhand to a convincing Syed Mushtaq Ali title triumph and felt that captaincy had helped him become a better batter.

Alcaraz crushes Paul, runs into Shelton in quarterfinal

US OPEN

Agence France-Presse
NEW YORK

Defending champion Carlos Alcaraz posted his most convincing win in his injury return at the US Open, downing Tommy Paul in straight sets on Sunday to line up a quarterfinal clash with Ben Shelton. Second-seeded Alcaraz looked closer to his brilliant best in a 6-4, 6-3, 6-4 victory over Paul, the 20th seed. Alcaraz will face red-hot Shelton, a 6-2, 6-3, 6-4 winner over Stefanos Tsitsipas, for a place in the semifinals. In the women's section, Linda Noskova reached the quarterfinals, edging Ukraine's Marta Kostyuk in three sets. The Czech, who has largely flown under the radar in New York despite winning Wimbledon, beat

his quarterfinal berth with a 7-6(1), 6-4, 7-6(6) victory over former champion Daniil Medvedev. Tiafoe will face compatriot Alex Michelsen, who defeated Argentina's Tomas Martin Etcheverry 7-6(6), 6-4, 6-4, for a place in the semifinals. Women's third seed Jessica Pegula, one of the players with a chance to supplant Sabalenka atop the rankings in New York, reached the last eight with a 6-3, 6-4 victory over Romanian Sorana Cirstea. On Monday, Paris Olympics gold medallist Zheng Qinwen, who had to come through qualifying, stunned six-time Grand Slam champion Iga Swiatek 7-5, 6-3 to reach the quarterfinals. Fifth-seeded French Open champion Mirra Andreeva beat Anastasia Potapova 5-7, 6-4, 6-3 to seal a last-eight berth.



Stroll in the park: Shelton proved too good for Tsitsipas. AP



Fit and raring: Satwik and Chirag say they are in good shape ahead of the Asian Games. BADMINTONPHOTO

‘It's always Plan A’ for Satwik-Chirag

BADMINTON

Anirudh Velamuri

Satwiksairaj Rankireddy and Chirag Shetty, down a game and staring at a third runner-up finish, tore up the script. They stormed back to beat home favourites He Ji Ting and Ren Xiang Yu 11-21, 21-13, 21-17 on Sunday, avenging two painful final losses in 2023 and 2025 to finally win the China Masters. Satwik credited the win to a marked improvement in their physical conditioning, coming just weeks after a disappointing quarterfinal exit at the badminton World championships at home. “We weren't sure how our bodies could cope throughout the week in China. But we became much better after the World championships. Our bodies settled down, and we got better. That's a good sign going forward,” Satwik said during a virtual interaction on Monday. Chirag added that the physical grind of their run in China had doubled as ideal preparation – mentally and physically – for the

final itself. Through it all, the attack-minded duo remained committed to their game plan. “It's always Plan A; there is no Plan B for us,” Satwik emphasised. “The Chinese want to play a fast game, and if you're playing with Malaysians and Indonesians, they're not as attacking,” he explained, while also crediting his partner for the pair's attacking edge. Their 10th crown on the BWF World Tour gives them a timely confidence boost heading into their Asian Games title defence later this month. Meanwhile, Chirag, spotted with a belt strapped around his waist during the post-match celebrations, was quick to wave off any fitness concerns ahead of the continental event. “I've been wearing it since the injury at the All England last year. If I get a period of one month, probably in December, when I can have a good stretch of practice, that is when I can try playing without it and see how I feel. Not confident enough until then,” he stressed.

Knights beat Pipers despite Anand's loss

GLOBAL CHESS LEAGUE

P.K. Ajith Kumar
BENGALURU

Viswanathan Anand rarely plays these days: he is more of a mentor and now the interim president of FIDE. Magnus Carlsen, who dethroned Anand as the World champion in 2013, has relinquished his crown. Still, when the two all-time greats come across a chessboard, there is a buzz. That was quite palpable at the Lalit Ashok on Monday. They played some interesting chess too, on the third day of the Tech Mahindra Global Chess League, but Anand was unlucky, as he lost on time from an equal position. The five-time World champion, however, could afford to forget his defeat, as his team, PBG Alaskan Knights, won the match against Alpine APL Pipers 9-7. That, in fact, was the first loss in the tournament for Carlsen's team after winning both their matches. A little later, however, the defending champion bounced back, scoring a crushing 15-3 win against Fyers American Gambits. Carlsen was held to a draw by Javokhir Sindarov, but K. Humpy, Vidit Gujrathi and Volodar Murzin won their

games, with black pieces, to ensure that their team would score heavily. The Pipers top the table after four rounds. It has nine match points, and 43 game points. As many as four teams are on six points – Mumbai Masters (36 game points), American Gambits (34), Alaskan Knights (30) and Ganges Grandmasters (28). Two-time champion Triveni Continental Kings, which finally managed a victory, has three points.

The results: PBG Alaskan Knights bt Alpine APL Pipers 9-7 (Anand lost to Carlsen; Arjun drew with Anish; Yagiz Kaan bt Vidit; Lagno drew with Divya; Batsiashvili bt Humpy; Abhimanyu drew with Murzin). Cheq Mumbai Masters bt Ganges Grandmasters 12-3 (Vachier-Lagrave drew with Nepomniachtch; Pranesh bt Aronian; Mamedyarov drew with Dominguez Perez; Harika drew with Shuvalova; Yip bt Tsolakidou; Abhimanyu bt Daneshvar). Fyers American Gambits lost to Alpine Pipers 3-15 (Sindarov drew with Carlsen; Nihal drew with Anish; Dubov lost to Vidit; Bibisara drew with Divya; Sarasadat lost to Humpy; Maurizzi lost to Murzin). Triveni Continental Kings bt Alaskan Knights 11-6 (Firouzja drew with Anand; Yi bt Arjun; Aravindh bt Yagiz Kaan; Jiner lost to Lagno; Kosteniuk bt Batsiashvili; Pranav drew with Abhimanyu).



Clash of titans: Anand was involved in an interesting game with Carlsen before losing on time. SUDHAKARA JAIN



Carlsen. FILE PHOTO: ALLEN EGENUSE J.

Chess in India is massive, says Carlsen

P.K. Ajith Kumar
BENGALURU Ahead of the Global Chess League, Magnus Carlsen had joked about how his team, Alpine APL Pipers, had done during his absence last year and hoped that it would not do worse this time. He should be pleased now. Pipers have gone top of the table despite Carlsen not being at his best so far. He has won only one of his four games – against his old rival Viswanathan Anand, and that too on time – but his teammates K. Humpy and Vidit Gujrathi have been doing rather well. The two Indians are, in fact, the best performing players in the league at the moment, both winning three of their games. “The team is doing fantastic,” Carlsen said. “Three wins out of four is great. My performance has been a bit underwhelming, but since the team is doing well. I am happy. It is nice to see good crowds here. Chess in India is massive.”

CM
YK

R. Kaushik

The Cambridge Dictionary defines *introspection* as the *examination of and attention to your own ideas, thoughts, and feelings*.

Introspection must be a ruthlessly honest endeavour. It's not an exercise to impress others but to look inwards, reflect on what one is and what needs to be done to become the best version of oneself. It calls for taking ownership of one's actions, of accepting mistakes and errors, if any, and of zealously working towards eradicating those mistakes and making amends, if so required.

There is no better example in Indian cricket of the use of introspection as a self-betterment tool than Ishan Kishan, the pocket-sized dynamite who has embarked on a sensational journey of reinvention that has made him one of the hottest properties, if only in the white-ball ecosystem for the time being.

To say that, at the beginning of 2026, his international career was at a crossroads will be a mild understatement. The destructive left-hander had worked himself out of favour as he allowed frustration, if not petulance, to get the better of him.

Overlooked from all three international playing XIs in the interest of team requirements and balance despite having done little wrong, Kishan took a leave of absence midway through the tour of South Africa in 2023-24, citing the need for a mental reset.

In choppy waters

But when he refused to obey the dictates of the Board of Control for Cricket in India and made himself unavailable to play domestic cricket for Jharkhand despite practising for IPL 2024, which was imminent, he found himself in choppy waters. In an unequivocal warning, the BCCI stripped him of his central contract - as it did with current T20I captain Shreyas Iyer. The unsaid message was clear - don't take anything for granted.

Kishan had one of two paths to choose - wallow in self-pity, rail at the injustice of it all and allow himself to go to seed, or smell the coffee, accept that he had done himself no favours and wake up to the reality that if he aspired to represent the country again, he would have to not just produce the runs expected of him but to reiterate without any scope for doubt that he had learned his lesson.

The Jharkhandi did all that and more. He was outstanding in domestic cricket

across formats, not just batting with flair and keeping with assurance but also slipping effortlessly into the role of a leader. His tactical acumen revealed itself time and again, manifesting in a maiden domestic title - the 20-over Syed Mushtaq Ali Trophy - last season with the captain himself starring in the final with a brazen century.

As India grappled with opening woes ahead of the home T20 World Cup, where they would be defending the title won in Bridgetown under Rohit Sharma in June 2024,

Kishan suddenly emerged as a very obvious and viable option. That the mandarins occupying the corridors of power had forgiven him became obvious when Suryakumar Yadav, the skipper, called up the younger man and asked him if he would win the team the World Cup.

Clearly, it wasn't a call Suryakumar made on his own. He would have consulted his bosses before approaching Kishan during a brief but very pithy chat. "*Chhotu, World Cup jitayega?*" the captain asked the wicketkeeper-batter, who responded with, "*Bharosa karogey?*" (Will you trust me?). Suryakumar's reply was instantaneous - "*Chal, kiya*" (Done). And just like that, Kishan was back in favour, back in business, back doing what he loves the most.

Celebrating the reintegration

Needless to say, Kishan kept up his end of the bargain. He celebrated his reintegration with international cricket with scores of 76, 28 and 103 in his second, third and fourth outings back, against New Zealand in India's final engagement before the World Cup. All these knocks came at No. 3, but with Sanju Samson suddenly blowing cold, he was thrust into the role of the opener-wicketkeeper at the World Cup, where he reeled off 20, 61, 77, 18 and 0 in the first five fixtures.

However, India were unable to get off to meaningful starts as opponents started to exploit the all-left opening salvo with off-spinners of varying quality.

Abhishek Sharma was in the middle of a horror run and therefore to break up the monotony of two left-handers, the think-tank took a punt on Samson, brought him back as opener and stumper and pushed Kishan down to No. 3 from the Super Eight fixture against Zimbabwe onwards. Samson came to the party with breathtaking knocks in the last three matches, including in the semis against England and the final against New Zealand, but Kishan didn't seem the drop

Listening to Kishan speak over the past few months is an eye-opening experience. One can see the deep impact of losing the central contract and finding himself in the dumps, but one can also easily make out that he has now become a more mature, grounded and realistic human being

Kishan 2.0 – a sensational reinvention

Having made giant strides, the southpaw, who struck a double-hundred in the Duleep final, could not have asked for a better start to the First Class season



R.V. MOORTHY

to one-down with 38 (24b), 10 (6), 39 (18) and a whirlwind 25-ball 54 in the title clash.

Chhotu did help Suryakumar lift the World Cup, but while his fortunes have continued to be on the upswing, the man who created history by becoming the first to oversee a successful title-retention campaign was immediately dumped without so much as a thank you.

His T20I heroics have brought Kishan back into the ODI mix as well. In June, he marked his comeback with a frenetic 125, off a mere 79 deliveries, against Afghanistan in Lucknow, in what was just his second 50-over international in 32 months. By this time, he had led Sunrisers Hyderabad for the first half of IPL 2026, with regular skipper Pat Cummins travelling with the squad but not cleared to enter the fray by Cricket Australia.

Even though he knew that he was only keeping the seat warm for the Australian Test and ODI skipper, Kishan wasn't a stand-in captain in the strictest sense. He brought his tactical acumen and man-management skills into play, led from the front and left the team in a good space - and handily placed to make a charge for the playoffs - when the Aussie reassumed charge.

Cummins was quick at every opportunity to laud Kishan's leadership skills in his absence, their mutual respect apparent from the repeated discussions the nominated captain had with the man who had led in his absence when the two were reunited on the field of play for the second half of the season.

Distinct and stated desire

Kishan isn't merely content with white-ball laurels, though. At least among Indian batters, there is a distinct and stated desire to represent the country in

Test cricket. Tilak Varma, the gifted 23-year-old, said as much on the eve of the ongoing Duleep Trophy final, where he is leading South Zone against Kishan's East.

Tilak has made a name for himself as an exuberant, new-age T20 player, a reputation that has landed him the vice-captaincy of the national side, and it won't be long before he becomes a 50-over certainty too. But he is consumed by the ambition to play for the country in the five-day format.

Since debuting in the Ranji Trophy for Hyderabad when he was only 16 years old, the rangy left-hander has stacked up an average of 55.77 from 24 First Class matches. He has often been unable to represent his state in the red-ball game because invariably, India's players are involved in some T20 outing or the other during the Ranji Trophy, but it's only a matter of time before Tilak gatecrashes the Test side as well.

In that race, he has Kishan for company and competition. Obviously, the shorter man is considerably older than Tilak and has greater experience of first-class cricket - 66 matches before the Duleep final, an average of just under 40 and nine hundreds with a best of 273. Kishan's red-ball quest is anything but straightforward.

The effervescent Rishabh Pant is the designated wicketkeeper-batter who is relishing his second coming at the international level after a long break following his terrible car accident in December 2022, while Dhruv Jurel has stolen a significant march as the gloveman-in-waiting.

Jurel's Test debut came when Pant was recuperating from a series of surgeries, and he immediately caught the eye with his agility and compactness behind the sticks and his composure and technique

in front of it. In only his second Test, he steered the team to a tense victory in Ranchi in the company of current skipper Shubman Gill and has stacked up two centuries in his last seven Tests.

How highly he is rated by the leadership group is evidenced by the fact that he is in the Test XI as a specialist batter, occupying the No. 6 position so that he can split left-handers Pant and Ravindra Jadeja.

His last Test knock was a flowing unbeaten 115 in Colombo a fortnight back and even though he is widely regarded as a terrific player of spin, it is hard to see him being dislodged when India play their next Test, in seam-friendly Wellington in November.

Work cut out

Kishan knows therefore that he has his work cut out, but no one also knows better than him that hard work can take you places you had never imagined were accessible. His statement-making double-hundred in the Duleep final in Chepauk is exactly the kind of start to the domestic First Class season he would have envisaged, and one can rest assured that when he is not on India white-ball duty, he will torment bowlers whilst masterminding Jharkhand's fortunes in a leadership capacity.

Listening to Kishan speak over the last few months is a refreshing, eye-opening experience. One can see the deep impact of losing the central contract and finding himself in the dumps, but one can also easily make out that he has now become a more mature, grounded and realistic human being who has still retained his innate impishness and the slant towards mischief. Clarity of thought is evident every time he opens his mouth, a clarity that is a spillover from his awareness in the middle.

In Ishan Kishan's fall from grace and his willingness to not just climb back on the horse but become a champion jockey again is a shining example of the wonders self-belief and commitment to the cause can catalyse. The intelligent young man is no longer chasing shadows or slaying inner demons. More power to him.

THE GIST

Kishan's T20I heroics have brought him back into the ODI mix. In June, he marked his comeback with a frenetic 125 against Afghanistan in Lucknow, in what was just his second 50-over international in 32 months

The southpaw isn't merely content with white-ball laurels, though. He knows that he has his work cut out, but no one knows better than him that hard work can take you places you had never imagined

In recent times, one can see the deep impact of the batter losing the central contract and finding himself in the dumps, but one can also easily make out that he has now become a more mature, grounded and realistic human being



R. RAGU



EMMANUAL YOGINI